These are notes for Malek Abdesselam's Functional Analysis class for Fall 2026.

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Motivation

Consider a space V of (sufficiently nice) functions $f : \mathbb{R} \to \mathbb{R}$. Then, similar to how a vector in $\mathbb{R}^n$ can be denoted as

$$x = \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix},$$

we may view the function $f : \mathbb{R} \to \mathbb{R}$ as a "column vector" with uncountable index. That is, $f = (f(x))_{x \in \mathbb{R}}$. If $L : V \to V$ is a linear map, then an analogue of the finite-dimensional matrix representation can be viewed as a kernel that f is integrated against; that is,

$$(Lf)(x) = \int_{\mathbb{R}} K(x,y) f(y)\, dy.$$

In the special case that $L = \mathrm{id}_V$, then the kernel $K(x,y)$ is given by the Dirac delta distribution, $K(x,y) = \delta(x-y)$. The space of these distributions is the primary object of study in Fourier analysis.

Notation

- The natural numbers are given by $\mathbb{N} = \{0, 1, 2, 3, \ldots\}$.
- The strictly positive natural numbers are given by $\mathbb{Z}_{>0} = \{1, 2, 3, \ldots\}$.
- A multi-index is a tuple $\alpha \in \mathbb{N}^n$. We let

$$|\alpha| = \alpha_1 + \cdots + \alpha_n.$$

If $x = (x_1, \dots, x_n) \in \mathbb{R}^n$, then we will write

$$x^\alpha = x_1^{\alpha_1} \cdots x_n^{\alpha_n}$$

and

$$\alpha! = \alpha_1! \cdots \alpha_n!$$

for the corresponding values.

If $f : \mathbb{R}^n \to \mathbb{R}$ is any function, then we will write

$$\partial^\alpha f = \frac{\partial^{|\alpha|}}{\partial x_1^{\alpha_1} \cdots \partial x_n^{\alpha_n}}.$$

Operations and comparators on multi-indices will be taken componentwise.

- If A is a set, we will write $\mathcal{P}(A)$ for the power set of A. If A and B are sets, then the set $B^A \subseteq \mathcal{P}(A \times B)$ is the set of all functions $f : A \to B$.

  We will write $\mathcal{P}_{\text{fin}}(X)$ to consist of all finite subsets subsets of X.

- If $x \in \mathbb{R}^n$ is given by $(x_1, \dots, x_n)$, we will define the bracket

$$\begin{aligned}\langle x \rangle &:= \sqrt{1 + x_1^2 + \cdots + x_n^2} \\ &= \sqrt{1 + |x|^2}.\end{aligned}$$

  In the particular case of $n = 1$, we will write $\langle k \rangle = \sqrt{1 + k^2}$ for any $k \in \mathbb{N}$. Note here that $|\cdot|$ denotes the norm on Euclidean space.

- If $n \in \mathbb{N}$, then

$$[n] = \{1, 2, \dots, n\},$$

  where we have $[0] = \emptyset$.

- If A is a finite set, then $|A|$ is the cardinality of A.
- Generally we will denote by $\mathbb{K}$ for either $\mathbb{R}$ or $\mathbb{C}$.
- If $V_1, \dots, V_n, W$ are topological vector spaces, we will write $\mathcal{L}_n(V_1, \dots, V_n; W)$ to denote the space of (jointly) continuous multilinear maps $V_1 \times \cdots \times V_n \to W$. If $n = 1$, we will just write $\mathcal{L}(V, W)$.
- The Borel σ-algebra on a topological space $(X, \mathcal{T}_X)$ is given by $\mathcal{B}_X$.
- If X is a compact metric space, the space $C(X)$ is the Banach space of continuous functions with values in the assumed field.

Locally Convex Topological Vector Spaces

Recall that if X is a set, then $\mathcal{T} \subseteq \mathcal{P}(\mathcal{P}(X))$ is a topology on X if it satisfies:

- $\emptyset, X \in \mathcal{T}$;
- $U_1, U_2 \in \mathcal{T}$ implies $U_1 \cap U_2 \in \mathcal{T}$;
- $\{U_i : i \in I\} \subseteq \mathcal{T}$ implies $\bigcup_{i \in I} U_i \in \mathcal{T}$.

Definition: Definitions

We say $(V, \mathcal{T})$ is a *topological vector space* (over $\mathbb{K}$) if V is a $\mathbb{K}$-vector space whose topology is such that the maps

$$\begin{aligned} V \times V &\to V \\ (u, v) &\mapsto u + v \end{aligned}$$

and

$$\begin{aligned} \mathbb{K} \times V &\to V \\ (\lambda, v) &\mapsto \lambda v \end{aligned}$$

are continuous with respect to the product topology. Note that $\mathbb{K}$ is equipped with its traditional topology.

The morphisms of topological vector spaces are the *continuous* linear maps. We say $V \cong W$ are isomorphic as topological vector spaces if there is $\varphi\colon V \to W$ that is bijective, linear, and continuous, and φ^{-1} is also continuous.

Definition: Let V be a $\mathbb{K}$-vector space. We say $\rho\colon V \to [0, \infty)$ is a *seminorm* if

$$\begin{aligned} \rho(v + w) &\leq \rho(v) + \rho(w) \\ \rho(\lambda v) &= |\lambda| \rho(v) \end{aligned}$$

for any $v, w \in V$ and $\lambda \in \mathbb{K}$.

Remark: The only difference between a seminorm and a norm is that $\rho(v) = 0$ need not imply that $v = 0$. A norm is a seminorm.

We will let $\mathrm{SN}(V)$ be the set of all seminorms of V.

Definition: Let V be a $\mathbb{K}$-vector space. Let $\mathcal{N} \subseteq \mathrm{SN}(V)$. Define $\mathcal{T}(\mathcal{N})$ to be a topology with basic open sets $U(x, F, \varepsilon) \subseteq V$ given by

$$\begin{aligned} U(x, F, \varepsilon) &= \{y \in V : \rho(y - x) < \varepsilon \text{ for all } \rho \in F\} \\ &= \bigcap_{\rho \in F} U(x, \rho, \varepsilon). \end{aligned}$$

for any $x \in V$, $F \subseteq \mathcal{N}$ finite, and $\varepsilon > 0$.

Remark: I prefer to use the subbasic open sets

$$U(x, \rho, \varepsilon) = \{y \in V : \rho(y - x) < \varepsilon\}$$

and write intersections.

We observe that the subsets of the form $U(x, \{\rho\}, \varepsilon)$ as ρ ranges across $\mathcal{N}$ forms a subbase for the topology of V.

Proposition: If $F \subseteq \mathcal{N}$ is finite, $\varepsilon > 0$, and $x \in V$, then

$$U(x, F, \varepsilon) \in \mathcal{T}(\mathcal{N}).$$

Proof. Suppose $y \in U(x, F, \varepsilon)$. Our goal is to find $\eta > 0$ such that $U(y, F, \eta) \subseteq U(x, F, \varepsilon)$. Suppose $z \in U(y, F, \eta)$. If $\rho \in F$, then

$$\begin{aligned} \rho(z - x) &\leq \rho(z - y) + \rho(y - x) \\ &< \eta + \rho(y - x). \end{aligned}$$

Therefore, we will define η to be small enough such that $\eta + \max_{\rho \in F} \rho(y - x) < \varepsilon$. This gives the desired result. □

Definition: A topological space is called a locally convex topological vector space (locally convex topological vector space) if the topology is generated by a family of seminorms $\mathcal{N}$ with the property that

$$\bigcap_{\rho\in\mathcal{N}}\{v : \rho(v) = 0\} = \{0\}.$$

A family $\mathcal{N}$ of seminorms that satisfies this property is said to be *separating*.

Example: If H is a Hilbert space with orthonormal basis $\{e_n : n \in \mathbb{N}\}$, then the seminorms $v \mapsto |\langle v, e_i\rangle|$ are separating.

Theorem: If V is a $\mathbb{K}$-locally convex topological vector space and $\mathcal{N}$ is a collection of seminorms that defines the topology on V, then the pair $(V, \mathcal{T}(\mathcal{N}))$ is Hausdorff if and only if $\mathcal{N}$ separates the points of V.

Proof. Assume $\mathcal{N}$ separates the points of V. Suppose $v \neq w$ are distinct elements of V. Then, since $v - w \neq 0$, there is some $\rho \in \mathcal{N}$ such that $\rho(v-w) = \varepsilon_0 > 0$. Consider the finite set $F = \{\rho\}$, and choose some $0 < \varepsilon < \varepsilon_0/2$. Then, we claim that the basic open sets $U(v, F, \varepsilon)$ and $U(w, F, \varepsilon)$ are disjoint. This follows from the fact that if $x \in U(v, F, \varepsilon)$, then $\varepsilon_0 = \rho(v-w) \leq \rho(v-x) + \rho(x-w)$, so that $\rho(w-x) \geq \varepsilon_0 - \varepsilon > \varepsilon_0 - \varepsilon_0/2 > \varepsilon$, yielding the desired result.

Now, suppose $\mathcal{N}$ does not separate the points of V. Then, there are some distinct $v \neq w$ such that for all $\rho \in \mathcal{N}$, we have $\rho(v-w) = 0$. In particular, observe that for all finite subsets $F \subseteq \mathcal{N}$ and all $\varepsilon > 0$, we have that w is contained in $U(v, F, \varepsilon)$, meaning that there are no basic open sets that separate v and w. Since any open subset of V contains a basic open subset, it follows that for any open subsets $v \in U$ and $w \in U'$, we have some finite F such that $U(v, F, \varepsilon) \subseteq U$, but this also has $w \in U(v, F, \varepsilon)$, implying that $U \cap U' \neq \emptyset$. Thus, V is not Hausdorff. □

Examples

Remark: If we remove the assumption that the family $\mathcal{N}$ is separating, then the topology induced by the trivial seminorm $\rho\colon V \to [0, \infty)$ is the indiscrete topology on V. Similarly, there is a family of seminorms that induces the discrete topology on V.

(i) Let $(V, \|\cdot\|)$ be a normed vector space over $\mathbb{K}$. Examples of these include Banach spaces such as $L^p(X, \Sigma, \mu)$, where Σ is a σ-algebra and μ is a positive measure on the measurable space (X, Σ), and $p \in [1, \infty]$.

There is a metric on this normed space $d(x, y) = \|x - y\|$, which gives rise to the metric topology $\mathcal{T}(d)$. We have $U \in \mathcal{T}(d)$ if and only if $U \subseteq V$ is such that for any $x \in U$, there exists $\varepsilon > 0$ such that

$$\begin{aligned}\{y \in V : d(x, y) < \varepsilon\} &=: U(x, \varepsilon)\\ &\subseteq U.\end{aligned}$$

This is a locally convex space with seminorm $\mathcal{N} = \{\|\cdot\|\}$. There are two possibilities for finite subsets $F \subseteq \{\|\cdot\|\}$, meaning that if $F = \emptyset$, then $U(x, F, \varepsilon) = U(x, \emptyset, \varepsilon) = V$, while if $F = \{\|\cdot\|\}$, then

$$\begin{aligned}U(x, F, \varepsilon) &= \{y \in V : \rho(x-y) < \varepsilon \text{ for all } \rho \in F\}\\ &= \{y \in V : \|x - y\| < \varepsilon\}\\ &= U(x, \varepsilon).\end{aligned}$$

(ii) Define $s(\mathbb{N}, \mathbb{K})$ to be the space of sequences $x = (x_n)_{n\geq 0}$ in $\mathbb{K}^{\mathbb{N}}$ such that for all $k \geq 0$, $\sup_{n\geq 0}\langle n\rangle^k |x_n| < \infty$, where $\langle n\rangle = \sqrt{1+n^2}$ is the bracket. This is the space of *rapidly decaying sequences* in $\mathbb{K}$. In essence, elements of $s(\mathbb{N}, \mathbb{K})$ decay faster than any inverse power of the index n.

We define the seminorms

$$\rho_k(x) = \sup_{n\geq 0}\langle n\rangle^k |x_n|.$$

In fact, the $\{\rho_k : k \in \mathbb{N}\}$ are norms. Let

$$\mathcal{N} := \{\rho_k : k \in \mathbb{N}\}$$

We obtain a locally convex on $s(\mathbb{N}, \mathbb{K})$ with topology induced by the family $\mathcal{N}$.

Later, we will see that $(s(\mathbb{N}, \mathbb{K}), \mathcal{T}(\mathcal{N}))$ is *not* a normed space. That is, there does not exist a norm $\|\cdot\|$ such that $\mathcal{T}(\mathcal{N}) = \mathcal{T}(\{\|\cdot\|\})$.

(iii) Take $V = C([0,\infty), \mathbb{R})$ to be the set of all real-valued continuous functions on $[0,\infty)$ with pointwise addition and scalar multiplication. This is a real vector space.

At first, we may think $\|f\| = \sup_{t\in\mathbb{R}}|f(t)|$ would define a norm, but this is not satisfactory since the functions in V are not necessarily bounded.

Therefore, we want to restrict to compact subsets. Take

$$\rho_k(f) := \sup_{0\le t\le k} |f(t)|.$$

Notice that the ρ_k are not norms (use Urysohn's Lemma or your favorite piecewise function to construct one). However, the family $\mathcal{N} = \{\rho_k : k \ge 0\}$ is a separating family.

Consider the set of functions $V_0 = \{f \in V : f(0) = 0\}$. Then, $V_0 \le V$ is a closed subset.

> *Proof.* Consider the linear functional $\mathrm{ev}_0 \colon V \to \mathbb{R}$ given by $f \mapsto f(0)$. We claim that this linear functional is continuous.
>
> Let $U \subseteq \mathbb{R}$ be open. Let $z \in U$, and let $\varepsilon > 0$ be such that $U(z,\varepsilon) := \left\{y \in \mathbb{R} : \left|z - y\right| < \varepsilon\right\}$ is contained in U. Let $f \in \mathrm{ev}_0^{-1}(U(z,\varepsilon))$, and set $\varepsilon_0 = |z - f(0)|$. To show that $\mathrm{ev}_0^{-1}(U(z,\varepsilon))$ is open, it suffices to find some seminorm ρ_k and some $\delta > 0$ such that $U(f, \rho_k, \delta) \subseteq \mathrm{ev}_0^{-1}(U(z,\varepsilon))$.
>
> In fact, we may choose $k = 1$ and $\delta = \varepsilon - \varepsilon_0$. Suppose $g \in U(f, \rho_1, \delta)$. Then, we have that
>
> $$\begin{aligned} |g(0) - z| &\le |g(0) - f(0)| + |f(0) - z| \\ &\le \sup_{t\in[0,1]} |g(t) - f(t)| + \varepsilon_0 \\ &< \delta + \varepsilon_0 \\ &= \varepsilon. \end{aligned}$$
>
> In particular, Thus, $U(f, \rho_1, \delta) \subseteq \mathrm{ev}_0^{-1}(U(z,\varepsilon))$, so that $\mathrm{ev}_0^{-1}(U(z,\varepsilon))$ is open as any element in this set admits a basic open neighborhood contained in the set. Therefore, ev_0 is continuous.
>
> Since $V_0 = \mathrm{ev}_0^{-1}(\{0\})$, it follows that V_0 is closed since $\{0\} \subseteq \mathbb{R}$ is closed. □

We claim that the subspace topology on V_0 is given by $\mathcal{T}_0 = \mathcal{T}\left(\left\{\rho_k|_{V_0} : k \ge 0\right\}\right)$.

> *Proof.* First, we will show that, given a subbasic open neighborhood $U = V_0 \cap U(x, \rho_k, \varepsilon)$ in the subspace topology of V_0 and any element y in this subbasic open neighborhood, there is an open neighborhood $V \in \mathcal{T}\left(\left\{\rho_k|_{V_0} : k \ge 0\right\}\right)$ such that such that $y \in V \subseteq U$.
>
> If $y \in U$, we observe that $y \in U(x, \rho_k, \varepsilon)$. In particular, since these form a subbase for $\mathcal{T}(\{\rho_k : k \ge 0\})$, it follows that there is some finite collection $F = \left\{\rho_{i_j} : j = 1, \dots, m\right\}$ of seminorms and some $\delta > 0$ such that
>
> $$U(y, F, \delta) = \bigcap_{j=1}^{m} U\left(y, \rho_{i_j}, \delta\right).$$

Since $U\left(y, \rho_{i_j}|_{V_0}, \delta\right) \subseteq U\left(y, \rho_{i_j}, \delta\right)$ for each j, it follows that we have

$$\bigcap_{j=1}^{m} U\left(y, \rho_{i_j}|_{V_0}, \delta\right) \subseteq V_0 \cap \left(\bigcap_{j=1}^{m} U\left(y, \rho_{i_j}, \delta\right)\right),$$

whence setting V to equal the left hand side yields the desired result. Therefore, $\mathcal{T}\left(\left\{\rho_k|_{V_0} : k \geq 0\right\}\right)$ refines the subspace topology $\mathcal{T}_0$.

In the reverse direction, observe that if $y \in U\left(x, \rho_j|_{V_0}, \varepsilon\right)$ in V_0, then $y \in U\left(x, \rho_j, \varepsilon\right)$, so there is some $\delta > 0$ and finite $F = \left\{\rho_{i_j} : j = 1, \ldots, m\right\}$ such that

$$\begin{aligned} U(y, F, \varepsilon) &= \bigcap_{j=1}^{m} U\left(y, \rho_{i_j}, \delta\right) \\ &\subseteq U\left(x, \rho_j, \varepsilon\right), \end{aligned}$$

but we notice then that $U\left(x, \rho_j, \varepsilon\right) \cap V_0 \subseteq U\left(x, \rho_j|_{V_0}, \varepsilon\right)$, so that $V_0 \cap U(y, F, \varepsilon) \subseteq U\left(x, \rho_j|_{V_0}, \varepsilon\right)$, so that $\mathcal{T}_0$ refines $\mathcal{T}\left(\left\{\rho_k|_{V_0} : k \geq 0\right\}\right)$. Thus, these topologies are equal. □

We may consider $\mathcal{B}(\mathcal{T}_0)$ to be the Borel σ-algebra on V_0. Brownian motions are a special class of probability measure on $(V_0, \mathcal{B}(\mathcal{T}_0))$. Specifically, they are the probability measures such that for every $n \geq 0$, partition $0 < t_1 < t_2 < \cdots < t_n$, the random variables

$$\begin{aligned} X_1 &= f \mapsto f(t_1) \\ X_2 &= f \mapsto (f(t_2) - f(t_1)) \\ X_3 &= f \mapsto (f(t_3) - f(t_2)) \\ &\vdots \\ X_n &= f \mapsto (f(t_n) - f(t_{n-1})) \end{aligned}$$

are independent centered Gaussian random variables with variances $t_1, t_2 - t_2, \ldots, t_n - t_{n-1}$ for an arbitrary f.

Similar to the case of $s(\mathbb{N}, \mathbb{K})$, the space $(V_0, \mathcal{T}_0)$ is not a normed space.

(iv) Consider the vector space $V = \mathbb{R}^{\mathbb{N}}$, which is the space of all real-valued sequences. Let $\mathcal{T}$ be the product topology on $\mathbb{R}^{\mathbb{N}}$. We are curious as to whether $(V, \mathcal{T})$ is a locally convex topological vector space.

Consider the function $d\colon V \times V \to [0, \infty)$ defined by

$$d(x, y) = \sum_{n=0}^{\infty} 2^{-n} \min\left(1, \left|x_n - y_n\right|\right).$$

Then, we have that d is a distance metric, and $\mathcal{T} = \mathcal{T}(d)$.

Proof. We start by showing that if $x \in V$ is any element and $\varepsilon > 0$, then $U(x, \varepsilon)$ is open in $\mathcal{T}$. Translating, we take $x = 0$ (since the metric d is translation-invariant), so that we are now seeking to show that $U(0, \varepsilon)$ is open. This is equivalent to showing that if $x \in U(0, \varepsilon)$, then there is a finite collection of subbasic open sets $\pi_j^{-1}\left(U_j\right)$ whose intersection W contains x and is contained in $U(0, \varepsilon)$.

To start, we observe that there is some index n_0 such that

$$\sum_{n=n_0+1}^{\infty} 2^{-n} < \varepsilon/2.$$

We then select $U_j = \left(x_j - \varepsilon/4, x_j + \varepsilon/4\right)$ for each $j = 0, \dots, n_0$. It then follows that $x \in W$, and we have for any $y \in W$ that

$$\begin{aligned} d(x,y) &= \sum_{j=0}^{\infty} 2^{-j} \min\left(1, \left|x_j - y_j\right|\right) \\ &\leq \sum_{j=0}^{n_0} 2^{-j} \min\left(1, \left|x_j - y_j\right|\right) + \sum_{j=n_0+1}^{\infty} 2^{-j} \\ &< \sum_{j=0}^{n_0} 2^{-j}(\varepsilon/4) + \varepsilon/2 \\ &\leq \varepsilon, \end{aligned}$$

whence $x \in W \subseteq U(0, \varepsilon)$.

Now, consider a basic open set $W = \bigcap_{j=0}^{n_0} \pi_j^{-1}\left(U_j\right)$. Let $x \in W$. Let $\delta_0, \dots, \delta_{n_0}$ be such that $\left(x_j - \delta_j, x_j + \delta_j\right) \subseteq U_j$ for each $j = 1, \dots, n$, and set $\delta = 2^{-n_1} \min\left(\delta_0, \dots, \delta_{n_0}\right)$, where n_1 is any integer strictly larger than n_0.

Notice then that if $y \in U(x, \delta)$, then we have for each $j = 0, \dots, n_0$ that

$$\begin{aligned} \left|x_j - y_j\right| &< 2^j \delta \\ &\leq 2^{n_0} 2^{-n_1} \min\left(\delta_0, \dots, \delta_{n_0}\right) \\ &< \min\left(\delta_0, \dots, \delta_{n_0}\right) \\ &\leq \delta_j. \end{aligned}$$

In particular, this means we must have $U(x, \delta) \subseteq W$.

Therefore, we have that $\mathcal{T}$ and $\mathcal{T}(d)$ generate the same topology on V. □

In particular, we may consider the seminorms

$$\rho_n(x) = |x_n|,$$

and $\mathcal{N} = \{\rho_n : n \geq 0\}$. Then, we claim that $\mathcal{T} = \mathcal{T}(\mathcal{N})$.

Proof. A basic open neighborhood of $x \in V$ in $\mathcal{T}(\mathcal{N})$ is given by the sets

$$U(x, F, \varepsilon) = \bigcap_{j \in F} \left\{(y_n)_n : \left|x_j - y_j\right| < \varepsilon\right\}.$$

Meanwhile, a basic open neighborhood of x in $\mathcal{T}$ is given by the sets

$$U'(x, F, \varepsilon) = \bigcap_{j \in F} \pi_j^{-1}\left(\left(x_j - \varepsilon, x_j + \varepsilon\right)\right).$$

In fact, these are the same subsets of V by the definition of the projection map $\pi_j \colon V \to \mathbb{R}$ taking $(x_n)_n \mapsto x_j$.

Therefore, if we have $y \in U(x, F, \varepsilon)$ in $\mathcal{T}(\mathcal{N})$, then if we define

$$\delta = \frac{1}{2} \min\left(\left\{\left|\varepsilon - y_j\right| : y_j \in F\right\}\right),$$

we have that $U'(y, F, \delta) \subseteq U(x, F, \varepsilon)$. Similarly, if $y \in U'(x, F, \varepsilon)$, then setting δ to equal the same as above, we have $U(y, F, \varepsilon) \subseteq U'(x, F, \varepsilon)$.

This gives $\mathcal{T} = \mathcal{T}(\mathcal{N})$. □

In particular, V is not a normed space.

(v) Consider $c_{00}(\mathbb{N},\mathbb{K}) \cong \mathbb{K}[x]$ to consist of all the finitely supported sequences $(x_n)_{n\geq 0}$. This is a vector space. Consider the seminorms $\mathcal{N} = \mathrm{SN}(V)$, and the locally convex space $(V,\mathcal{T}(\mathcal{N}))$. Then, this is a locally convex space that is not metrizable. The topology $\mathcal{T}(\mathcal{N})$ is known as the finest locally convex topology.

General Properties of Topological Vector Spaces

Definition: Let $(V,\mathcal{T})$ be a $\mathbb{K}$-TVS, and let $\tau \in \mathrm{SN}(V)$. Then, τ is a *continuous seminorm* for $(V,\mathcal{T})$ if $\tau\colon V \to [0,\infty) \subseteq \mathbb{R}$ is continuous.

Proposition: Let V be a $\mathbb{K}$-vector space, $\mathcal{N} \subseteq \mathrm{SN}(V)$ define a locally convex topology on V. If $\tau \in \mathrm{SN}(V)$, then τ is continuous for $(V,\mathcal{T}(\mathcal{N}))$ if and only if there exists $C \geq 0$, $n \in \mathbb{N}$, and $\rho_1,\dots,\rho_n \in \mathcal{N}$ such that

$$\tau(v) \leq C \max\Big\{\rho_j(v) : j = 1,\dots,n\Big\} \tag{$*$}$$

or

$$\tau(v) \leq C \sum_{j=1}^{n} \rho_j(v).$$

Proof. Suppose τ is continuous. Then, $[0,1)$ is open in $[0,\infty)$, so $\tau^{-1}([0,1))$ is open in $\mathcal{T}(\mathcal{N})$ with $0 \in \tau^{-1}([0,1))$.

So, by definition, there exists $\varepsilon > 0$ and $\rho_1,\dots,\rho_n$ such that

$$\bigcap_{j=1}^{n} U\Big(0,\rho_j,\varepsilon\Big) \subseteq \tau^{-1}([0,\infty)).$$

That is, for any $i \in [n]$, we have that if $\rho_i(x) < \varepsilon$, then $\tau(x) < 1$.

Let $x \in V$ and $\lambda > 0$. Then,

$$\begin{aligned}\tau(x) &= \tau\Big(\lambda\lambda^{-1}x\Big)\\ &= \lambda\tau\Big(\lambda^{-1}x\Big)\\ &< \lambda\end{aligned}$$

if for every i, we have $\rho_i\Big(\lambda^{-1}x\Big) < \varepsilon$.

Define $J = \Big\{\lambda > 0 : \rho_i\Big(\lambda^{-1}x\Big) < \varepsilon \text{ for all } i \in [n]\Big\}$. Rewriting, we have

$$\begin{aligned}J &= \left\{\lambda > 0 : \frac{1}{\varepsilon}\rho_i(x) < \lambda \text{ for all } i \in [n]\right\}\\ &= \bigcap_{i=1}^{n}\left(\frac{1}{\varepsilon}\rho_i(x),\infty\right),\end{aligned}$$

giving $\inf(J) = \frac{1}{\varepsilon}\max\{\rho_i(x) : 1 \leq i \leq n\}$. Therefore, if $\lambda \in J$, then $\tau(x) \leq \lambda$, whence $\tau(x) \leq \inf(J)$. Therefore, we have shown that $\tau(x) \leq \frac{1}{\varepsilon}\max\{\rho_i(x) : 1 \leq i \leq n\}$.

In the reverse direction, suppose $(*)$ holds. Let $x \in V$, $\rho_1,\dots,\rho_n \in \mathcal{N}$, and let $\eta > 0$. Let

$$y \in \bigcap_{i=1}^{n} U(x,\rho_i,\eta).$$

By the reverse triangle inequality, we have

$$\Big|\tau(y) - \tau(x)\Big| \leq \tau(y-x)$$

$$< \varepsilon,$$

but

$$\begin{aligned}\tau(y-x) &\leq C\max\{\rho_i(y-x) : 1 \leq i \leq n\} \\ &\leq C\eta \\ &< \varepsilon\end{aligned}$$

if η is small enough. Thus, we have for all $y \in \bigcap_{j=1}^{n} U\Big(x, \rho_j, \eta\Big)$, then $\Big|\tau(y) - \tau(x)\Big| < \varepsilon$, whence τ is continuous at x. □

Remark: If $\tau \in \mathcal{N}$, then τ is continuous with respect to $\mathcal{T}(\mathcal{N})$.

Proposition: Let $(V, \mathcal{T}(\mathcal{N}))$ be a locally convex topological vector space. Then, $U \subseteq V$ is open if and only if for all $x \in U$, there is some continuous seminorm τ and some $\varepsilon > 0$ such that $U(x, \tau, \varepsilon) \subseteq U$.

Theorem (Continuity of Multilinear Maps on locally convex topological vector spaces)**:** Suppose $n \geq 1$, and $V_1, \ldots, V_n, W$ are $\mathbb{K}$-locally convex topological vector spaces, and let $M\colon\ V_1 \times \cdots \times V_n \to W$ be a $\mathbb{K}$-multilinear map.

Then, M is (jointly) continuous if and only if for all continuous seminorm τ on W, there exist $\rho_1, \ldots, \rho_n$ continuous seminorms on $V_1, \ldots, V_n$ respectively such that for all tuples $(x_1, \ldots, x_n) \in V_1 \times \cdots \times V_n$,

$$\tau(M(x_1, \ldots, x_n)) \leq \rho(x_1) \cdots \rho(x_n). \tag{†}$$

Proof. Suppose M is (jointly) continuous. That is, M is equipped with the product topology. Since τ is a continuous seminorm, we have that τ is a continuous map from $W \to [0, \infty)$, we have $\tau^{-1}([0,1))$ is open in W. Moreover, $M(0, \ldots, 0) = 0 \in \tau^{-1}([0,1))$. Since M is continuous at $(0, \ldots, 0)$, there is a neighborhood $U_1 \times \cdots \times U_n$ in $V_1 \times \cdots \times V_n$ where each U_i is a neighborhood of 0 in V_i, such that $M(U_1 \times \cdots \times U_n) \subseteq \tau^{-1}([0,1))$.

Let $(x_1, \ldots, x_n) \in V_1 \times \cdots \times V_n$, and let $\lambda_1, \ldots, \lambda_n > 0$. We have

$$\begin{aligned}\tau(M(x_1, \ldots, x_n)) &= \tau\Big(M\Big(\lambda_1\lambda_1^{-1}x_1, \ldots, \lambda_n\lambda_n^{-1}x_n\Big)\Big) \\ &= \lambda_1 \cdots \lambda_n \tau\Big(M\Big(\lambda_1^{-1}x_1, \ldots, \lambda_n^{-1}x_n\Big)\Big) \\ &< \lambda_1 \cdots \lambda_n\end{aligned}$$

if for all i, we have $\lambda_i^{-1}x_i \in U_i$. For each $i \in [n]$, we have U_i is an open neighborhood of $0 \in V_i$, so there exists some continuous seminorm σ_i and $\varepsilon_i > 0$ such that $U(0, \sigma_i, \varepsilon_i) \subseteq U_i$. If for every i, we have $\sigma_i\Big(\lambda_i^{-1}x_i\Big) < \varepsilon_i$, then

$$\tau(M(x_1, \ldots, x_n)) < \lambda_1 \cdots \lambda_n.$$

Rewriting, we have $\lambda_i > \frac{1}{\varepsilon_i}\sigma_i(x_i)$ for each such i, so that we have

$$\begin{aligned}\tau(M(x_1, \ldots, x_n)) &\leq \lambda_1 \cdots \lambda_n \\ &\leq \inf\left(\left\{\lambda_1 \cdots \lambda_n : \lambda_i > 0 \text{ and for all } i \in [n],\ \lambda_i > \frac{1}{\varepsilon_i}\sigma_i(x_i)\right\}\right) \\ &= \left(\frac{1}{\varepsilon_1}\sigma_1(x_1)\right) \cdots \left(\frac{1}{\varepsilon_n}\sigma_n(x_n)\right)\end{aligned}$$

Setting $\rho_i = \frac{1}{\varepsilon_i}\sigma_i$ yields the desired result.

Assume that (†) holds. Let $(x_1, \ldots, x_n), (y_1, \ldots, y_n) \in V_1 \times \cdots \times V_n$. We want to try to compare the difference between $M(x_1, \ldots, x_n)$ and $M(y_1, \ldots, y_n)$. From multilinearity, we have

$$M(x_1, \ldots, x_n) - M(y_1, \ldots, y_n) = \sum_{i=1}^{n} (M(y_1, \ldots, y_{i-1}, y_i, x_{i+1}, \ldots, x_n) - M(y_1, \ldots, y_{i-1}, x_i, x_{i+1}, \ldots, x_n))$$

$$= \sum_{i=1}^{n} M(y_1, \dots, y_{i-1}, y_i - x_i, x_{i+1}, \dots, x_n)$$

Let $Z \subseteq W$ be open containing $z = M(x_1, \dots, x_n)$, and write $z' = M(y_1, \dots, y_n)$. Then, there exists a continuous seminorm τ on W and $\varepsilon > 0$ such that $U(z, \tau, \varepsilon) \subseteq Z$.

Consider $\tau(z - z')$. Then, we have by multilinearity and the triangle inequality that

$$\tau(z - z') \leq \sum_{i=1}^{n} \tau(M(y_1, \dots, y_{i-1}, y_i - x_i, x_{i+1}, \dots, x_n)).$$

Using (†) gives continuous seminorms $\rho_1, \dots, \rho_n$ on $V_1, \dots, V_n$ respectively such that

$$\begin{aligned} &\leq \sum_{i=1}^{n} \rho_1(y_1) \cdots \rho_{i-1}(y_{i-1}) \rho_i(y_i - x_i) \rho_{i+1}(x_{i+1}) \cdots \rho_n(x_n) \\ &\leq n(C + \eta)^{n-1} \eta, \end{aligned}$$

where we assume that for all i, $y_i \in U(x_i, \rho_i, \eta)$, and $C = \max\{\rho_i(x_i) : i = 1, \dots, n\}$. By choosing η sufficiently small, $n(C + \eta)^{n-1} \eta < \varepsilon$, which yields

$$\begin{aligned} M\left(\prod_{i=1}^{n} U(x_i, \rho_i, \eta) \right) &\subseteq U(z, \tau, \varepsilon) \\ &\subseteq Z, \end{aligned}$$

so M is continuous. □

Theorem (Refinement of Previous Theorem)**:** Let $V_1, \dots, V_n$ and W be locally convex topological vector spaces, and let $\mathcal{N}$ be a defining collection of seminorms for the topology on W. Then, if $M : V_1 \times \cdots \times V_n \to W$ is a multilinear map, we have M is continuous if and only if for every $\tau \in \mathcal{N}$, there exist continuous seminorms $\rho_1, \dots, \rho_n$ on $V_1, \dots, V_n$ respectively such that for all $x = (x_1, \dots, x_n)$,

$$\tau(M(x)) \leq \prod_{i=1}^{n} \rho_i(x_i).$$

Proof. The forward direction follows from the fact that τ is a continuous seminorm, while the reverse direction follows from the fact that any continuous seminorm is dominated by a constant multiple of some family of seminorms in $\mathcal{N}$, and a constant multiple of the continuous seminorm ρ_i on V_i is also a continuous seminorm on V_i. □

Corollary: If $T : V \to W$ is a linear map of locally convex topological vector spaces $\mathcal{N}$ defining the topology on W, then T is continuous if and only if for all $\tau \in \mathcal{N}$ (or all continuous seminorms on W), there exists a continuous seminorm ρ on V such that for all $x \in V$, $\tau(T(x)) \leq \rho(x)$.

Proposition: Let V be a $\mathbb{K}$-vector space, and let $\mathcal{N}_1, \mathcal{N}_2 \subseteq \mathrm{SN}(V)$. Then, $\mathcal{T}(\mathcal{N}_1) \subseteq \mathcal{T}(\mathcal{N}_2)$ if and only if for all $\tau \in \mathcal{N}_1$, τ is a continuous seminorm for the locally convex topological vector space $(V, \mathcal{T}(\mathcal{N}_2))$.

Proof. We have that $\mathcal{T}(\mathcal{N}_1)$ is contained in $\mathcal{T}(\mathcal{N}_2)$ if and only if the identity map $\mathrm{id} : (V, \mathcal{T}(\mathcal{N}_2)) \to (V, \mathcal{T}(\mathcal{N}_1))$ is continuous. □

Remark: We recover the fact that $(V, \|\cdot\|_1)$ and $(V, \|\cdot\|_2)$ are equivalent as norms if and only if the identity map $\mathrm{id} : (V, \|\cdot\|_1) \to (V, \|\cdot\|_2)$ is a homeomorphism.

Normability and Metrizability

We will focus on Hausdorff locally convex topological vector spaces from now on. That is, the defining family of seminorms separates points.

Definition: Let $(V, \mathcal{T}(\mathcal{N}))$ be a Hausdorff locally convex topological vector space. We say V is normable if and only if there exists a norm $\|\cdot\|$ on V such that $\mathcal{T}(\mathcal{N}) = \mathcal{T}(\|\cdot\|)$.

Theorem: Let $(V, \mathcal{T}(\mathcal{N}))$ be a locally convex topological vector space. The following are equivalent:

(i) $(V, \mathcal{T}(\mathcal{N}))$ is normable;

(ii) there exists a finite subset F of $\mathcal{T}(\mathcal{N})$ such that $\mathcal{T}(\mathcal{N}) = \mathcal{T}(F)$;

Proof. We start by showing that (i) implies (ii). Let $\|\cdot\|$ be a norm such that $\mathcal{T}(\mathcal{N}) = \mathcal{T}(\|\cdot\|)$. Since $\mathcal{T}(\|\cdot\|) \subseteq \mathcal{T}(\mathcal{N})$, there exist $\rho_1, \ldots, \rho_n \in \mathcal{N}$ and some constant C such that

$$\|\cdot\| \leq C \max\{\rho_k : k = 1, \ldots, n\}.$$

Let $\tau \in \mathcal{N}$. Then, since $\mathcal{T}(\mathcal{N}) \subseteq \mathcal{T}(\|\cdot\|)$, it follows that there is some K such that $\tau \leq K\|\cdot\|$, whence

$$\tau \leq CK \max\{\rho_k : k = 1, \ldots, n\}$$

In particular, this means that the finite subset of seminorms $F = \{\rho_1, \ldots, \rho_n\}$ generates the topology $\mathcal{T}(\mathcal{N})$ since every seminorm in $\mathcal{N}$ is continuous with respect to $\mathcal{T}(F)$.

Next, we show that (ii) implies (i). Suppose there is a finite subset $F \subseteq \mathcal{N}$ with elements $\rho_1, \ldots, \rho_n$ such that $\mathcal{T}(\mathcal{N}) = \mathcal{T}(\mathcal{F})$. In particular, we have that for all seminorms $\tau \in \mathcal{N}$, we have

$$\tau \leq C \max\{\rho_k : k = 1, \ldots, n\}.$$

We claim that

$$\|\cdot\| = \max\{\rho_k : k = 1, \ldots, n\}$$

induces the topology on $\mathcal{N}$. This follows from the fact that every seminorm $\tau \in \mathcal{N}$ is continuous with respect to $\|\cdot\|$, so $\mathcal{T}(\mathcal{N}) \subseteq \mathcal{T}(\|\cdot\|) \subseteq \mathcal{T}(\mathcal{N})$, with the latter inclusion emerging from the fact that $\|\cdot\|$ is necessarily a continuous seminorm in $\mathcal{N}$ by the way it was defined. Since $(V, \mathcal{T}(\mathcal{N}))$ is Hausdorff, it follows that $\|\cdot\|$ separates points, so $\|\cdot\|$ is a norm. □

This allows us to prove that certain examples we discussed earlier are not normable spaces.

Example: Consider

$$V = \mathbb{R}^{\mathbb{N}}$$

with topology defined by seminorms $\rho_n = |x_n|$. Then, we observe that there are no finite subsets of the seminorms $\mathcal{N} = \{\rho_k : k \in \mathbb{N}\}$ that separate the points of V by virtue of the fact that if $F = \left\{\rho_{i_1}, \ldots, \rho_{i_j}\right\}$ is a finite subset, then the elementary vector e_{i_j+1} with 1 in position $\left(i_j + 1\right)$ is taken to 0 by any continuous seminorm on $\mathcal{T}(\mathcal{F})$, but e_{i_j+1} is not zero.

Next, we concern ourselves with metrizability of topological vector spaces.

Recall that (X, τ) is *first countable* if for any $x \in X$, the system of open neighborhoods $\mathcal{O}_x$ has a countable basis (i.e., a countable neighborhood basis). That is, there is some $\{U_n : n \geq 1\} \subseteq \mathcal{O}_x$ such that for any $Z \in \mathcal{O}_x$, there is some n such that $U_n \subseteq Z$. Metric spaces are always first countable by virtue of the Archimedean property.

Proposition: A (not necessarily locally convex) TVS $(V, \mathcal{T})$ is first countable if and only if 0 has a countable neighborhood basis.

Proof. Addition by a fixed vector $v \in V$ is a homeomorphism of V. □

Theorem: Let $(V, \mathcal{T}(\mathcal{N}))$ be a locally convex topological vector space. The following are equivalent:

(i) the space $(V, \mathcal{T}(\mathcal{N}))$ is metrizable;

(ii) the space $(V, \mathcal{T}(\mathcal{N}))$ is first countable;

(iii) there exists a countable subset $\mathcal{M} \subseteq \mathcal{N}$ such that $\mathcal{T}(\mathcal{M}) = \mathcal{T}(\mathcal{N})$.

Proof. We see that (i) implies (ii) since metrizable spaces are first countable.

Next, we show that (ii) implies (iii). Let $\{U_n : n \geq 0\}$ be a neighborhood basis for 0. Then, for any n, there exists finite $F_n \subseteq \mathcal{N}$ and $\varepsilon_n > 0$ such that

$$\bigcap_{\rho \in F_n} U(0, \rho, \varepsilon_n) \subseteq U_n.$$

Taking $\mathcal{M} = \bigcup_{n \geq 0} F_n$, this yields a countable subcollection of $\mathcal{N}$. We will show that $\mathcal{T}(\mathcal{N}) \subseteq \mathcal{T}(\mathcal{M})$. That is, we will have to show that for all $\tau \in \mathcal{N}$, τ is continuous with respect to $\mathcal{M}$.

We have $\tau^{-1}([0,1))$ is a neighborhood of 0 in $\mathcal{T}(\mathcal{N})$. This contains U_n for some $n \in \mathbb{N}$, and thus some finite subset F_n with

$$\begin{aligned} \bigcap_{\rho \in F_n} U(0, \rho, \varepsilon_n) &\subseteq U_n \\ &\subseteq \tau^{-1}([0,1)). \end{aligned}$$

In particular, this means that if $x \in V$ is such that for all $\rho \in F_n$, $\rho(x) < \varepsilon_n$, then $x \in \tau^{-1}([0,1))$, whence τ is continuous. Scaling allows one to apply the same method for any $x \in V$.

Finally, we will show that (iii) implies (i). Let $\mathcal{M} \subseteq \mathcal{N}$ be a countable subset with seminorms $\{\rho_n : n \geq 0\}$. Let

$$d(x,y) = \sum_{n=0}^{\infty} 2^{-n} \min\{\rho_n(x-y), 1\}.$$

Then, this is a distance metric and $\mathcal{T}(d) = \mathcal{T}(\mathcal{M})$. □

Definition: A locally convex topological vector space $(V, \mathcal{T})$. If $(V, \mathcal{T})$ is complete metrizable, then we call V a *Fréchet space*.

Example:

(i) Banach spaces are Fréchet spaces.

(ii) The space of rapidly decreasing sequences $s(\mathbb{N}, \mathbb{K})$ is a Fréchet space that is not normable. The distance metric is given by

$$d(x,y) = \sum_{k=0}^{\infty} 2^{-k} \min(1, \rho_k(x-y)).$$

(iii) The vector space $V = \mathbb{R}^{\mathbb{N}}$ with the product topology is a Fréchet space.

(iv) The space of finitely supported sequences $c_{00}(\mathbb{N}, \mathbb{K})$, henceforth denoted c_{00}, with the finest locally convex topology is a topological vector space that is not metrizable, let alone a Fréchet space. This follows from a diagonal argument on the assumption of countably many seminorms.

Consider the uncountable collection of seminorms $\mathcal{N}$ given by, for any $\omega \in [0,\infty)^{\mathbb{N}}$,

$$\rho_\omega(x) = \sum_{n=0}^{\infty} \omega_n |x_n|,$$

where we observe that each ρ_ω is distinct. It is possible to show that $\mathcal{T}(\mathcal{N}) = \mathcal{T}(\mathrm{SN}(c_{00}))$.

One direction follows directly from the definition. The reverse direction uses the fact that $\{e_n : n \geq 0\}$ forms an algebraic basis for $c_0 0$. If ρ is a seminorm, then we will define $\omega_n = \rho(e_n)$. For any $x \in c_{00}$,

then we may write

$$x = \sum_{n=0}^{N} x_n e_n,$$

satisfying

$$\begin{aligned}\rho(x) &= \rho\left(\sum_{n=0}^{N} x_n e_n\right) \\ &\leq \sum_{n=0}^{N} \omega_n |x_n| \\ &= \rho_\omega(x).\end{aligned}$$

Since x is arbitrary, it follows that every seminorm is continuous with respect to the seminorms of the family $\mathcal{N}$, whence the family $\mathcal{N}$ defines the topology on c_{00}.

In fact, we may assume that $\omega \in \mathbb{N}^{\mathbb{N}}$ by replacing $(\omega_n)_n \in [0,\infty)^{\mathbb{N}}$ with $(\lceil \omega_n \rceil)_n$.

Notable Examples

Example: Let $K \subseteq \mathbb{R}^d$ be a compact subset. Define

$$\mathcal{D}_K\left(\mathbb{R}^d\right) = C_K^\infty\left(\mathbb{R}^d\right)$$

to be the space of C^∞ with $\operatorname{supp}(f) \subseteq K$. Define seminorms for each $N \geq 0$ by

$$\rho_N(f) = \max_{\substack{\alpha \in \mathbb{N}^d \\ |\alpha| \leq N}} \sup_{x \in \mathbb{R}^d} \left|\partial^\alpha f(x)\right|.$$

Let $\mathcal{T} = \mathcal{T}(\{\rho_N : N \geq 0\})$. This is a countable collection of seminorms, so the space $C_K^\infty\left(\mathbb{R}^d\right)$ equipped with $\mathcal{T}$ is a Fréchet space with the metric

$$d(f,g) = \sum_{N \geq 0} 2^{-N} \min(1, \rho_N(f-g)).$$

Example: A function $f : \mathbb{R}^d \to \mathbb{C}$ is called a Schwartz function if $f \in C^\infty\left(\mathbb{R}^d\right)$ and for all $\alpha \in \mathbb{N}^d$ and all $k \in \mathbb{N}$,

$$\sup_{x \in \mathbb{R}^d} \langle x \rangle^k \left|\partial^\alpha f(x)\right| < \infty.$$

We label these seminorms by $\rho_{\alpha,k}(f)$. The space $\mathcal{S}\left(\mathbb{R}^d\right)$ of all such functions is known as the *Schwartz space* for $\mathbb{R}^d$.

The standard topology on $\mathcal{S}\left(\mathbb{R}^d\right)$ is given by $\mathcal{T} = \mathcal{T}(\mathcal{N})$, where $\mathcal{N}$ is the collection of all such $\rho_{\alpha,k}$. We may also write $\|\cdot\|_{\alpha,k}$ for $\rho_{\alpha,k}$

We can also consider the family of all such ρ_N with

$$\rho_N(f) = \max_{\substack{|\alpha| \leq N \\ k \leq N}} \rho_{\alpha,k}(f),$$

which yields the metric

$$d(f,g) = \sum_{N=0}^{\infty} 2^{-N} \min(1, \rho_N(f-g)).$$

This induces the same topology as the family of all the $\rho_{\alpha,k}$.

With this distance metric, the Schwartz space is complete. In particular, this means that $\mathcal{S}\left(\mathbb{R}^d\right)$ is an example of a Fréchet space. The Schwartz space is needed to define the Fourier transform (which we will discuss more heavily later). The Fourier transform is defined by

$$\hat{f}(\xi) = \int_{\mathbb{R}^d} e^{-ix\xi} f(x)\, d^d x.$$

When $f \in L^1$, we have that $\hat{f}$ is well-defined, but $\hat{f}$ is not necessarily in L^1. It turns out that the Fourier transform is an automorphism of the Schwartz space (over $\mathbb{C}$). Note that Schwartz functions are L^1 since they decay faster than functions of the form

$$u(x) = \langle x \rangle^{\lambda}$$

where $\lambda < d$. That such a $u(x)$ is integrable can be found by using the hyperspherical coordinates.

Example (A Non-Example)**:** The dual space of the Schwartz space, $\mathcal{S}\left(\mathbb{R}^d\right)^*$, is called the space of *tempered distributions*. We say $\mathcal{S}\left(\mathbb{R}^d\right)^* \subseteq C_c^\infty\left(\mathbb{R}^d\right)^*$, where the dual space $C_c^\infty\left(\mathbb{R}^d\right)^*$ is called the space of all distributions. We will discuss these more soon.

The space $C_c^\infty\left(\mathbb{R}^d\right)$ is the space of compactly supported C^∞ functions. In particular,

$$\begin{aligned} C_c^\infty\left(\mathbb{R}^d\right) &= \bigcup_{K \text{ compact}} C_K^\infty\left(\mathbb{R}^d\right) \\ &= \bigcup_{N \geq 1} C_{B(0,N)}^\infty\left(\mathbb{R}^d\right), \end{aligned}$$

where

$$B(0,N) = \left\{x \in \mathbb{R}^d : |x| \leq N\right\}$$

is the closed ball centered at 0.

Though $C_c^\infty\left(\mathbb{R}^d\right)$ is the increasing countable union of Fréchet spaces, it is not itself Fréchet; in fact, $C_c^\infty\left(\mathbb{R}^d\right)$ is not even metrizable. It is an example of an inductive limit of a family of Fréchet spaces.

Let

$$\iota_K \colon C_K^\infty\left(\mathbb{R}^d\right) \to C_c^\infty\left(\mathbb{R}^d\right)$$

denote the inclusion maps. We say a seminorm ρ on $C_c^\infty\left(\mathbb{R}^d\right)$ is compatible if and only if for every compact $K \subseteq \mathbb{R}^d$, $\rho \circ \iota_K$ is a continuous seminorm on $C_K^\infty\left(\mathbb{R}^d\right)$. The topology on $C_c^\infty\left(\mathbb{R}^d\right)$ is the topology generated by all compatible seminorms.

Definition: We say a sequence $(f_n)_n$ of functions on $U \subseteq \mathbb{R}^d$ converges locally uniformly to g if for all $x \in U$, there is an open subset $V \subseteq U$ containing x such that

$$\lim_{n \to \infty} \sup_{y \in V} \left|f_n(y) - g(y)\right| = 0.$$

Theorem: The sequence $(f_n)_n$ converges locally uniformly to g if and only if $(f_n)_n$ converges uniformly on compact subsets.

Proposition: Let $(f_n)_n$ be a sequence of C^1 functions on $\mathbb{R}^d$, and suppose f_n converges pointwise to $f \colon U \to \mathbb{C}$. Furthermore, assume that for all $i \in [d]$, $\partial_i f_n$ converges locally uniformly to some function $g_i \colon U \to \mathbb{C}$. Then, f is C^1 and $g_i = \partial_i f$ for all i.

Proof. Since each g_i is the local uniform limit of some continuous functions, each g_i is continuous. Fix $x \in U$.

Since $\partial_i f_n \to g_i$ locally uniformly, there is some $\varepsilon > 0$ such that

$$\sup_{y \in U(x,\varepsilon)} \left|\partial_i f_n(y) - g_i(y)\right| \xrightarrow{n\to\infty} 0.$$

For some $t \in \mathbb{R}$ with $0 < |t| < \varepsilon$, the fundamental theorem of calculus gives

$$f_n(x + te_i) - f_n(x) = t \int_0^1 \partial_i f_n(x + ste_i)\, ds.$$

Taking the limit as $n \to \infty$ and using dominated convergence gives

$$\begin{aligned} f(x + te_i) - f(x) &= \lim_{n\to\infty} t \int_0^1 \partial_i f_n(x + ste_i)\, ds \\ &= t \int_0^1 g_i(x + ste_i)\, ds. \end{aligned}$$

Therefore, we have

$$\begin{aligned} \left|\frac{f(x + te_i) - f(x)}{t} - g_i(x)\right| &= \left|\int_0^1 g_i(x + ste_i) - g(x)\, ds\right| \\ &\leq \sup_{0\leq s\leq 1} |g_i(x + ste_i) - g_i(x)| \\ &\to 0 \end{aligned}$$

by the continuity of g_i at x. Thus, $\partial_i f = g_i$, so f is C^1. □

Duality

Let $(V, \mathcal{T})$ be a topological vector space. The algebraic dual space V' is the space of all linear functionals $L\colon V \to \mathbb{C}$. The topological dual space, V^*, is the space of all continuous linear functionals $L\colon V \to \mathbb{C}$.

Definition: The weak* topology on V^* is the topology induced by the family of seminorms

$$\mathcal{N}_{\text{wk}^*} := \{(L \mapsto |L(x)|) : x \in V\}.$$

There is another locally convex topology on V^* that is the analogue of the norm topology in the case of a normed vector space. To understand this analogue, we must discuss boundedness.

Definition: Let $(V, \mathcal{T})$ be a topological vector space. We say A is bounded if for all open neighborhoods $U \in \mathcal{O}_0$ of 0, there is $\lambda > 0$ such that $\lambda A \subseteq U$.

Theorem: Let V be a normed vector space. Then, A is bounded in the above definition if and only if A is bounded in the usual sense.

Proposition: Let $(V, \mathcal{T})$ be a topological vector space, and let $A \subseteq V$. The following are equivalent:

(i) A is bounded;

(ii) for all $\rho \in \mathcal{N}$, $\sup_{x\in A} \rho(x) < \infty$;

(iii) for all continuous seminorms ρ, $\sup_{x\in A} \rho(x) < \infty$.

Proof. Assume (i); that is, suppose A is bounded, and let ρ be a defining seminorm. Then, there exists some $\lambda > 0$ such that $\lambda A \subseteq U(0, \rho, 1)$. That is, for all $x \in A$,

$$\lambda \rho(x) < 1,$$

or that $\rho(x) < \lambda^{-1}$. Therefore, $\sup_{x\in A} \rho(x) \leq \lambda^{-1}$, giving (ii).

Suppose (ii) holds. Let ρ be a continuous seminorm on $(V,\mathcal{T})$. Then, we may write $\rho(x) \leq C\max\{\rho_i(x) : i = 1,\dots,n\}$ for some family of defining seminorms ρ_i. This gives (iii).

Assume (iii) holds. Let $U \in \mathcal{O}_0$ be an open neighborhood of 0. Then, there exists some continuous seminorm ρ and some $\varepsilon > 0$ such that $U(0,\rho,\varepsilon) \subseteq U$. Notice that for any $x \in A$ and any $\lambda > 0$, we have

$$\begin{aligned}\rho(\lambda x) &= \lambda\rho(x)\\ &\leq \lambda \sup_{y\in A}\rho(y).\end{aligned}$$

Since $M = \sup_{y\in A}\rho(y)$ is finite by assumption, choosing λ sufficiently small such that $0 < \lambda < \varepsilon/M$ yields $\rho(\lambda x) < \varepsilon$ for all $x \in A$, or that $\lambda A \subseteq U(0,\rho,\varepsilon)$. □

Theorem: Let $(V,\mathcal{T})$ be a topological vector space. Then, $\mathcal{T}$ is a norm topology if and only if V contains a bounded nonempty open neighborhood of 0.

Definition: Let $(V,\mathcal{T})$ be a topological vector space. The *strong dual* is the topology on V^* defined by the family of all ρ_A with A bounded, where

$$\rho_A(L) = \sup_{x\in A}|L(x)|.$$

Definition: Let V and W be topological vector spaces with strong duals V^* and W^* respectively. Let $\varphi\colon V \to W$ be a continuous linear map. Then, the transpose is given by $\varphi^*\colon W^* \to V^*$ taking $\varphi^*(L) = L\circ\varphi$.

Proposition: The transpose of a continuous linear map is well-defined, linear, and continuous.

Proof. Our goal is to show that, given $L \in V^*$, we have that $\varphi^*(L) = L\circ\varphi$ is a continuous linear functional. This follows from the fact that L and φ are continuous. Similarly, the linearity of φ^* follows directly from definition.

To show that φ^* is continuous, the seminorm ρ_A on V^* for any $A \subseteq V$ bounded. Then, for any $L \in W^*$, we have

$$\begin{aligned}\rho_A(\varphi^*(L)) &= \sup_{x\in A}\big|L\circ\varphi(x)\big|\\ &= \sup_{y\in\varphi(A)}\big|L(y)\big|\\ &= \rho_{\varphi(A)}(L).\end{aligned}$$

We note that $\varphi(A)$ is itself bounded. □

Fourier Analysis

The next section of our coverage is focused on Fourier analysis.

Schwartz Distributions

Recall that the dual space $C_c^\infty\left(\mathbb{R}^d\right)^*$ of compactly supported C^∞ functions is called the space of *distributions*, while the dual space $\mathcal{S}\left(\mathbb{R}^d\right)^*$ is called the space of *tempered distributions*.

Given $\theta \in \mathcal{S}\left(\mathbb{R}^d\right)^*$ and $f \in \mathcal{S}\left(\mathbb{R}^d\right)$, we write $\langle f,\theta\rangle = \theta(f)$ (this will be held more generally for dual systems too). Since $C_c^\infty\left(\mathbb{R}^d\right) \subseteq \mathcal{S}\left(\mathbb{R}^d\right)$, we will define a canonical inclusion map $\varphi\colon C_c^\infty\left(\mathbb{R}^d\right) \to \mathcal{S}\left(\mathbb{R}^d\right)$.

Theorem: The inclusion φ is continuous with respect to the locally convex topology on each space.

Proof. Once again we use the seminorm criterion. Let $\alpha \in \mathbb{N}^d$ and $k \in \mathbb{N}$. Consider the seminorm $\|\cdot\|_{\alpha,k}$ on $\mathcal{S}$. We will show that there exists a continuous seminorm ρ on $C_c^\infty\left(\mathbb{R}^d\right)$ such that $\left\|\varphi(f)\right\|_{\alpha,k} \leq C\rho(f)$ for all $f \in C_c^\infty\left(\mathbb{R}^d\right)$.

It would suffice to show that for any compact $K \subseteq \mathbb{R}^d$, $\|\cdot\|_{\alpha,k} \circ \iota_K$ is a continuous seminorm on $C_c^\infty(K)$, where $\iota_K \colon C_c^\infty(K) \subseteq C_c^\infty\left(\mathbb{R}^d\right)$ is the inclusion. Indeed, we have

$$\begin{aligned}\|\iota_K(f)\|_{\alpha,k} &= \sup_{x\in\mathbb{R}^d} \langle x\rangle^k \left|\partial^\alpha \iota_K(f)(x)\right| \\ &= \sup_{x\in K} \langle x\rangle^k \left|\partial^\alpha f(x)\right| \\ &\leq \left(\sup_{x\in K}\langle x\rangle^k\right)\sup_{x\in K}\left|\partial^\alpha f(x)\right|.\end{aligned}$$

Defining $C = \sup_{x\in K}\langle x\rangle^k$, we have that $C < \infty$ by compactness, so we obtain our desired result. □

Theorem: The space of test functions, $C_c^\infty\left(\mathbb{R}^d\right)$, is sequentially dense in the Schwartz space.

Proof. We start by defining a smooth bump function $\varphi \colon \mathbb{R}^d \to [0,1]$ such that $\varphi(x) = 1$ for any $x \in [-1,1]^d$ and $\varphi(x) = 0$ for all $x \in \mathbb{R}^d \setminus (-2,2)^d$. Take $\rho_1, \rho_2 \colon \mathbb{R} \to \mathbb{R}$ by taking

$$\rho_1(x) = \begin{cases} 0 & x \leq 0 \\ e^{-1/x} & x > 0 \end{cases}$$
$$\rho_2(x) = \rho_1(x)\rho_1(1-x).$$

Then, ρ_1 is smooth, so ρ_2 is smooth as well. Moreover, $\rho_2(x) = 0$ for any $x \notin (0,1)$, and $\rho_2(x) \geq 0$ For any $x \in (0,1)$. Let

$$C = \int_0^1 \rho_2(x)\,dx$$
$$\rho_3(x) = \frac{1}{C}\int_0^x \rho_2(y)\,dy.$$

Then, $\rho_3(x) = 1$ for all $x \geq 1$, $\rho_3(x) = 0$ for all $x \leq 0$, and it is smooth in between. If we let ρ_4 be given by $\rho_4(x) = \rho_3(2-x)\rho_3(2+x)$, then for any $1 \leq x \leq 1$, we have $\rho_3(2-x) = \rho_3(2+x) = 1$, and $\rho_4(x) = 0$ for any $|x| \geq 2$, with ρ_4 otherwise smooth and bounded between 0 and 1. Finally, let $\varphi \colon \mathbb{R}^d \to [0,1]$ be given by

$$\varphi = \rho_4(x_1)\cdots\rho_4(x_d).$$

Let $f \in \mathcal{S}\left(\mathbb{R}^d\right)$. Let $\varphi_n(x) = \varphi\left(\frac{x}{n}\right)$, and let $f_n(x) = f(x)\varphi_n(x)$. Our goal is to show that $f_n \to f$ in the topology of $\mathcal{S}\left(\mathbb{R}^d\right)$. Let $\alpha \in \mathbb{N}^d$ and $k \in \mathbb{N}$. We must show that

$$\begin{aligned}\|f - f_n\|_{\alpha,k} &= \sup_{x\in\mathbb{R}^d}\langle x\rangle^k\left|\partial^\alpha f_n(x) - \partial^\alpha f(x)\right| \\ &\xrightarrow{n\to\infty} 0.\end{aligned}$$

Note that for any $x \in (-n,n)^d$, we have $|\partial^\alpha f_n(x) - \partial^\alpha f(x)| = 0$ since the truncation and the original function are equal on $[-n,n]^d$. Thus, we only need to consider $x \notin (-n,n)^d$. By the Leibniz rule, we have

$$\begin{aligned}\partial^\alpha f_n(x) &= \partial^\alpha(\varphi_n(x)f(x)) \\ &= \sum_{\substack{\beta,\gamma\in\mathbb{N}^d \\ \beta+\gamma=\alpha}} \frac{\alpha!}{\beta!\gamma!}\partial^\beta\varphi_n(x)\partial^\gamma f(x).\end{aligned}$$

Applying chain rule gives

$$\partial^\alpha f_n(x) = \sum_{\substack{\beta,\gamma\in\mathbb{N}^d \\ \beta+\gamma=\alpha}} \frac{\alpha!}{\beta!\gamma!} n^{-|\beta|}\left(\partial^\beta\varphi\right)\left(\frac{x}{n}\right)\partial^\gamma f(x).$$

Since the $\gamma = \alpha$ term is eliminated, we get

$$\partial^\alpha f_n(x) - \varphi_n(x)\partial^\alpha f(x) = \sum_{\substack{\beta,\gamma\in\mathbb{N}^d \\ \beta+\gamma=\alpha, |\beta|\geq 1}} \frac{\alpha!}{\beta!\gamma!} n^{-|\beta|}\Big(\partial^\beta \varphi\Big)\Big(\frac{x}{n}\Big)\partial^\gamma f(x)$$

Therefore, we obtain

$$\Big|\partial^\alpha f_n(x) - \varphi_n(x)\partial^\alpha f(x)\Big| \leq \sum_{\substack{\beta,\gamma\in\mathbb{N}^d \\ \beta+\gamma=\alpha, |\beta|\geq 1}} \frac{\alpha!}{\beta!\gamma!} \big\|\varphi\big\|_{\beta,0} \langle x\rangle^{-k-1} \|f\|_{\gamma,k+1}.$$

Choose some constant C_1 greater than or equal to the right hand side, so that we have

$$\langle x\rangle^k \Big|\partial^\alpha f_n(x) - \varphi_n(x)\partial^\alpha f(x)\Big| \leq C_1 \langle x\rangle^{-1}.$$

Therefore, by the triangle inequality, we have

$$\langle x\rangle^k \Big|\partial^\alpha f_n(x) - \partial^\alpha f(x)\Big| \leq C_1\langle x\rangle^{-1} + \langle x\rangle^k \Big|\varphi_n(x)\partial^\alpha f(x) - \partial^\alpha f(x)\Big|.$$

Since $0 \leq \varphi_n(x) \leq 1$, we have

$$\begin{aligned}\langle x\rangle^k \Big|\varphi_n(x)\partial^\alpha f(x) - \partial^\alpha f(x)\Big| &\leq \langle x\rangle^k \Big(\langle x\rangle^{-k-1}\|f\|_{\alpha,k+1}\Big)\Big|\varphi_n(x) - 1\Big| \\ &\leq \langle x\rangle^{-1}\|f\|_{\alpha,k+1},\end{aligned}$$

so that

$$\langle x\rangle^k \Big|\partial^\alpha f_n(x) - \partial^\alpha f(x)\Big| \leq \frac{C + \|f\|_{\alpha,k+1}}{\langle x\rangle}.$$

Since for any $x \notin (-n,n)^d$, we have $\langle x\rangle \geq |x| \geq n$, it follows that

$$\begin{aligned}\|f_n - f\|_{\alpha,k} &= \sup_{x\in\mathbb{R}^d} \langle x\rangle^k \Big|\partial^\alpha f_n(x) - \partial^\alpha f(x)\Big| \\ &\leq \frac{C_1 + \|f\|_{\alpha,k}}{\langle x\rangle} \\ &\leq \frac{C_1 + \|f\|_{\alpha,k+1}}{n} \\ &\xrightarrow{n\to\infty} 0.\end{aligned}$$

□

Corollary: The space of tempered distributions, $\mathcal{S}\big(\mathbb{R}^d\big)^*$ is contained in the space of distributions $C_c^\infty\big(\mathbb{R}^d\big)^*$.

Proof. We have shown that the inclusion map $\varphi\colon C_c^\infty\big(\mathbb{R}^d\big) \to \mathcal{S}\big(\mathbb{R}^d\big)$ is linear and continuous, so the transpose $\varphi^*\colon \mathcal{S}\big(\mathbb{R}^d\big)^* \to C_c^\infty\big(\mathbb{R}^d\big)^*$ is linear and continuous. Our goal is to show the injectivity of the transpose map.

Yet, this follows from the fact that if $L \in \mathcal{S}\big(\mathbb{R}^d\big)^*$ is such that $\varphi^*(L) = 0$, then $L(f) = 0$ for all $f \in C_c^\infty\big(\mathbb{R}^d\big)$, but since L is continuous and $C_c^\infty\big(\mathbb{R}^d\big) \subseteq \mathcal{S}\big(\mathbb{R}^d\big)$ is sequentially dense, we have that $L = 0$ so φ^* is injective. □

Classical Fourier Transform

We will assume that $\mathbb{K} = \mathbb{C}$.

Definition: Let

$$\begin{aligned}\mathcal{F}\colon \mathcal{S}\left(\mathbb{R}^d,\mathbb{C}\right) &\to \mathcal{S}\left(\mathbb{R}^d,\mathbb{C}\right)\\ f &\mapsto \hat{f} = \mathcal{F}[f]\end{aligned}$$

be defined by

$$\hat{f}(\xi) := \int_{\mathbb{R}^d} e^{-i\langle x,\xi\rangle} f(x)\, d^d x.$$

Remark: We will let

$$\mathcal{G}\colon \mathcal{S}\left(\mathbb{R}^d,\mathbb{C}\right) \to \mathcal{S}\left(\mathbb{R}^d,\mathbb{C}\right)$$

be defined by

$$\mathcal{G}\left[\hat{f}\right](x) = \int_{\mathbb{R}^d} e^{i\langle \xi,x\rangle} \hat{f}(\xi)\, d^d \xi.$$

Theorem: The map $\mathcal{F}$ is well-defined, linear, and continuous.

Proof. We recall the seminorms

$$\|f\|_{\alpha,k} = \sup_{x\in\mathbb{R}^d} \langle x\rangle^k \left|\partial^\alpha f(x)\right|.$$

We want to verify that $\mathcal{F}$ is well-defined. We note that

$$\begin{aligned}\int_{\mathbb{R}^d} \left|e^{-i\langle x,\xi\rangle} f(x)\right| d^d x &= \int_{\mathbb{R}^d} \left|\langle x\rangle^{-(d+1)} \langle x\rangle^{d+1} f(x)\right| d^d x\\ &\leq \|f\|_{0,d+1} \int_{\mathbb{R}^d} \langle x\rangle^{-(d+1)}\, d^d x\\ &< \infty.\end{aligned}$$

Our next job is to show that $\hat{f}$ is C^∞. By dominated convergence, we have that

$$\begin{aligned}\partial_j \hat{f}(\xi) &= \int_{\mathbb{R}^d} \frac{\partial}{\partial \xi_j}\left(e^{-i\langle x,\xi\rangle} f(x)\right) d^d x\\ &= \int_{\mathbb{R}^d} \left(-ix_j\right) e^{-i\langle x,\xi\rangle} f(x)\, d^d x.\end{aligned}$$

Dominated convergence applies since applying the derivative to the integrand yields the upper bound given by

$$\begin{aligned}\left|\left(-ix_j\right) e^{-i\langle x,\xi\rangle} f(x)\right| &\leq \sqrt{1 + x_1^2 + \cdots + x_n^2}|f(x)|\\ &= \langle x\rangle |f(x)|\\ &=: g(x),\end{aligned}$$

which is integrable since f is a Schwartz function.

Inductively, we have

$$\left|\frac{\partial^n}{\partial \xi_{j_1}\cdots\partial\xi_{j_1}} e^{-i\langle x,\xi\rangle} f(x)\right| \leq \langle x\rangle^n |f(x)|.$$

In general, for any multi-index $\alpha \in \mathbb{N}^d$, we have

$$\partial^\alpha \hat{f}(\xi) = \int_{\mathbb{R}^d} (-i)^{|\alpha|} x^\alpha e^{-i\langle x,\xi\rangle} f(x)\, d^d x.$$

Therefore, $\hat{f}$ is C^∞.

Our next goal is to extract the decay rate so that $\hat{f}$ is in Schwartz space. Let $\alpha, \beta \in \mathbb{N}^d$, and consider

$$\begin{aligned}\xi^\beta \partial_\xi^\alpha \hat{f}(\xi) &= \int_{\mathbb{R}^d} (-i)^{|\alpha|} x^\alpha \xi^\beta e^{-i\langle x,\xi\rangle} f(x)\, d^d x \\ &= i^{|\beta|-|\alpha|} \int_{\mathbb{R}^d} \partial_x^\beta \left(e^{-i\langle x,\xi\rangle}\right) x^\alpha f(x)\, d^d x,\end{aligned}$$

where we note that $\partial_x^\beta \left(e^{-i\langle x,\xi\rangle}\right) = i^{-|\beta|}\xi^\beta e^{-i\langle x,\xi\rangle}$.

Integrating by parts and the Leibniz rule yield

$$\begin{aligned}\xi^\beta \partial_\xi^\alpha \hat{f}(\xi) &= (-i)^{|\alpha|+|\beta|} \int_{\mathbb{R}^d} e^{-i\langle x,\xi\rangle} \partial_x^\beta (x^\alpha f(x))\, d^d x \\ &= (-i)^{|\alpha|+|\beta|} \int e^{-i\langle x,\xi\rangle} \sum_{\substack{0\le\gamma\le\beta \\ \gamma\in\mathbb{N}^d}} \frac{\beta!}{\gamma!(\beta-\gamma)!} \partial_x^\gamma (x^\alpha) \partial_x^{\beta-\gamma}(f(x))\, d^d x \\ &= (-i)^{|\alpha|+|\beta|} \sum_{\gamma\in\mathbb{N}^d} \chi_{\{\gamma\le\min(\alpha,\beta)\}} \frac{\alpha!\beta!}{\gamma!(\alpha-\gamma)!(\beta-\gamma)!} \int e^{-i\langle x,\xi\rangle}\, d^d x.\end{aligned}$$

Estimating yields

$$\left|\xi^\beta \partial_\xi^\alpha \hat{f}(\xi)\right| \le \sum_{\gamma\in\mathbb{N}^d} \chi_{\{\gamma\le\min(\alpha,\beta)\}} \frac{\alpha!\beta!}{\gamma!(\alpha-\gamma)!(\beta-\gamma)!} \left(\int_{\mathbb{R}^d} \langle x\rangle^{-(d+1)}\, d^d x\right) \|f\|_{\beta-\gamma,|\alpha|-|\gamma|+d+1}.$$

In particular, the right-hand side is a finite constant, so that

$$\begin{aligned}\sup_{\xi\in\mathbb{R}^d} \left|\langle\xi\rangle^{2m} \partial_\xi^\alpha \hat{f}(\xi)\right| &= \sup_{\xi\in\mathbb{R}^d} \left(1+\xi_1^2+\cdots+\xi_d^2\right)^m \left|\partial_\xi^\alpha \hat{f}(\xi)\right| \\ &< \infty.\end{aligned}$$

Thus, $\hat{f} \in \mathcal{S}\left(\mathbb{R}^d\right)$. Furthermore, this estimate yields that $\mathcal{F}$ is continuous, as any defining seminorm applied to $\hat{f}$ is bounded by a constant multiple of a seminorm applied to f.

Similarly, the map $\mathcal{G}$ simply substitutes i for $-i$, so that $\mathcal{G}$ is also continuous. □

The most important Fourier transform that we will compute is the Gaussian. That is, $f(x) = e^{-|x|^2/2}$.

Theorem: If $f(x) = e^{-|x|^2/2}$ over $\mathbb{R}^d$, we have $\hat{f}(\xi) = (2\pi)^{d/2} e^{-\xi^2/2}$.

Proof. Notice that $f(x) \in \mathcal{S}\left(\mathbb{R}^d\right)$. Fubini's theorem yields

$$\begin{aligned}\hat{f}(\xi) &= \int_{\mathbb{R}^d} e^{-i\langle x,\xi\rangle} e^{-|x|^2/2}\, d^d x \\ &= \int_{\mathbb{R}^d} \prod_{j=1}^d e^{-i\xi_j x_j - \left(x_j^2/2\right)}\, d^d x \\ &= \prod_{j=1}^d \int_{\mathbb{R}} e^{-|x|^2/2 - i\langle\xi,x\rangle}\, dx.\end{aligned}$$

Therefore, we reduce to the one-dimensional case. One can complete the square, yielding

$$\begin{aligned}\int_{-\infty}^{\infty} e^{-|x|^2/2-i\langle \xi,x\rangle}\,dx &= e^{-\xi^2/2}\int_{-\infty}^{\infty} e^{-\frac{1}{2}(x+i\xi)^2}\,dx \\ &= e^{-\xi^2/2}\lim_{A\to\infty}\int_{-A}^{A} e^{-\frac{1}{2}(x+i\xi)^2}\,dx\end{aligned}$$

Translating by $-i\xi$ and using the residue theorem on the resulting rectangular contour yields that the integral is the same as the original Gaussian integral.

The second method follows by a Taylor expansion. By dominated convergence, we have

$$\begin{aligned}\int_{-\infty}^{\infty} e^{-|x|^2/2-i\langle \xi,x\rangle}\,dx &= \int_{-\infty}^{\infty} x^n e^{-|x|^2/2}\sum_{n=0}^{\infty}\frac{(i|\xi|)^n}{n!}\,dx \\ &= \sum_{n=0}^{\infty}\frac{(i|\xi|)^n}{n!}\int_{-\infty}^{\infty} x^n e^{-|x|^2/2}\,dx.\end{aligned}$$

If X is a random variable with standard normal distribution, we note that the integral evaluates to

$$\int_{-\infty}^{\infty} x^n e^{-|x|^2}\,dx = (2\pi)^{1/2}\mathbb{E}[X^n],$$

while

$$\mathbb{E}[X^n] = \begin{cases} 0 & n \text{ is odd} \\ \frac{(2n)!}{2^n n!} & n \text{ is even.}\end{cases}$$

In particular, we have that

$$\begin{aligned}\int_{-\infty}^{\infty} e^{-|x|^2/2-i\langle \xi,x\rangle}\,dx &= (2\pi)^{1/2}\sum_{k=0}^{\infty}\frac{(-i)^{2k}\xi^{2k}}{(2k)!}\frac{(2k)!}{2^k k!} \\ &= (2\pi)^{1/2}e^{-\xi^2/2}.\end{aligned}$$

□

Our goal is to prove the Fourier Inversion Theorem.

Theorem (Fourier Inversion Theorem)**:** Let $\mathcal{F}$ and $\mathcal{G}$ be defined by

$$\begin{aligned}\mathcal{F}[f](\xi) &= \int_{\mathbb{R}^d} e^{-i\langle x,\xi\rangle} f(x)\,d^d x \\ \mathcal{G}\left[\hat{f}\right](x) &= \int_{\mathbb{R}^d} e^{i\langle x,\xi\rangle}\hat{f}(\xi)\,d^d\xi.\end{aligned}$$

Then, $\mathcal{G}\circ\mathcal{F} = \mathcal{F}\circ\mathcal{G} = (2\pi)^d$.

In particular, we obtain the following corollary.

Corollary: The map $\mathcal{F}$ is a topological vector space isomorphism from $\mathcal{S}\left(\mathbb{R}^d,\mathbb{C}\right)$ to itself, with inverse given by

$$\mathcal{F}^{-1}[f](x) = \frac{1}{(2\pi)^d}\int_{\mathbb{R}^d} e^{i\langle x,\xi\rangle} f(\xi)\,d^d\xi.$$

We will write $\check{f}(x)$ for $\mathcal{F}^{-1}[f](x)$.

To prove the Fourier Inversion Theorem, we will need the following proposition.

Proposition: We have

$$\lim_{\varepsilon \searrow 0} (2\pi\varepsilon)^{-d/2} e^{-\frac{|y-x|^2}{2\varepsilon}} = \delta_x^d(y)$$

in $\mathcal{S}\left(\mathbb{R}^d, \mathbb{C}\right)^*$ with the strong topology. Here, δ_x^d is the evaluation map implemented by the duality pairing

$$\left\langle f, \delta_x^d \right\rangle = \int_{\mathbb{R}^d} f(y)\delta_x^d(y)\, d^d y.$$

Remark: The duality pairing for $f \in \mathcal{S}\left(\mathbb{R}^d, \mathbb{C}\right)$ and $\varphi \in \mathcal{S}\left(\mathbb{R}^d, \mathbb{C}\right)^*$ written as

$$\langle f, \varphi \rangle = \int f(y)\varphi(y)\, dy$$

is not really an integral. However, physicists like writing it this way.

Proof of Proposition. Let $\varphi_{x,\varepsilon} \in \mathcal{S}\left(\mathbb{R}^d, \mathbb{C}\right)$ be defined by

$$\varphi_{x,\varepsilon}(y) = (2\pi\varepsilon)^{-d/2} e^{-\frac{|y-x|^2}{2\varepsilon}}.$$

There is a canonical injection $\iota \colon \mathcal{S}\left(\mathbb{R}^d, \mathbb{C}\right) \to \mathcal{S}\left(\mathbb{R}^d, \mathbb{C}\right)^*$.

Let $I = [0, \infty)$ be directed by taking $\varepsilon_1 \preceq \varepsilon_2$ if and only if $\varepsilon_2 \leq \varepsilon_1$. Then, our goal is to show that the net $(\iota(\varphi_{x,\varepsilon}))_{\varepsilon \in I}$ converges to δ_x^d in the strong topology.

We want to show that, for all bounded subsets $A \subseteq \mathcal{S}\left(\mathbb{R}^d, \mathbb{C}\right)$, one has

$$\begin{aligned} 0 &= \lim_{\varepsilon \searrow 0} \left\| \iota_{\varphi_{x,\varepsilon}} - \delta_x^d \right\|_A \\ &= \sup_{f \in A} \left| \left\langle f, \iota(\varphi_{x,\varepsilon}) - \delta_x^d \right\rangle \right|. \end{aligned}$$

We see that

$$\begin{aligned} \left\langle f, \iota(\varphi_{x,\varepsilon}) - \delta_x^d \right\rangle &= \int_{\mathbb{R}^d} \varphi_{x,\varepsilon}(y) f(y)\, d^d y - f(x) \\ &= \int_{\mathbb{R}^d} \varphi_{x,\varepsilon}(y)(f(y) - f(x))\, d^d y, \end{aligned}$$

where we note that $\varphi_{x,\varepsilon}$ was specifically chosen to have integral 1 over $\mathbb{R}^d$. Using the Fundamental Theorem of Calculus, we have

$$\begin{aligned} &= \int_{\mathbb{R}^d} \varphi_{x,\varepsilon}(y) \int_0^1 \langle y - x, \nabla f(x + t(y-x)) \rangle\, dt\, d^d y \\ &= \int_{\mathbb{R}^d} (2\pi\varepsilon)^{-d/2} e^{-\frac{|y-x|^2}{2}} \int_0^1 \langle (y-x), \nabla f(x + t(y-x)) \rangle\, dt\, d^d y \\ &= \sum_{j=1}^d \int_{\mathbb{R}^d} (2\pi\varepsilon)^{-d/2} e^{-\frac{|y-x|^2}{2\varepsilon}} \int_0^1 \left\langle y_j - x_j, \partial_j f(x + t(y-x)) \right\rangle dt\, d^d y. \end{aligned}$$

Estimating the inner integral yields

$$\left| \left\langle f, \iota(\varphi_{x,\varepsilon}) - \partial_x^d \right\rangle \right| \leq \sum_{j=1}^d \|f\|_{e_j,0} \int_{\mathbb{R}^d} \left| y_j - x_j \right| (2\pi\varepsilon)^{-d/2} e^{-\frac{|y-x|^2}{2\varepsilon}}\, d^d y$$

$$\leq d \max_{1\leq j\leq d} \int_{\mathbb{R}^d} (2\pi\varepsilon)^{-d/2} e^{-\frac{|y-x|^2}{2\varepsilon}} \left|y-x\right| d^d y.$$

Substituting $u = \frac{y-x}{\sqrt{\varepsilon}}$ yields $y = x + \sqrt{\varepsilon}u$, so that

$$\leq d\left(\max_{1\leq j\leq d} \sup_{f\in A} \|f\|_{e_j,0}\right)\sqrt{\varepsilon}\left(\int_{\mathbb{R}^d} (2\pi)^{-d/2} e^{-\frac{|u|^2}{2}} |u|\, d^d u\right).$$

Since the integral is finite, we obtain that $\left\|\iota(\varphi_{x,\varepsilon}) - \delta_x^d\right\|_A \leq C\sqrt{\varepsilon}$. □

Remark: The functions of the form $(2\pi\varepsilon)^{-d/2} e^{-\frac{|y-x|^2}{2\varepsilon}}$ are a special class of functions known as mollifiers. We will now prove the Fourier Inversion Theorem.

Proof of the Fourier Inversion Theorem. Let $f \in \mathcal{S}\left(\mathbb{R}^d, \mathbb{C}\right)$. We compute

$$\mathcal{G}\circ\mathcal{F}[f](x) = \int_{\mathbb{R}^d} e^{i\langle x,\xi\rangle} \hat{f}(\xi)\, d^d\xi.$$

Since $\hat{f}$ is a Schwartz function, dominated convergence gives

$$\begin{aligned}
&= \lim_{\varepsilon\searrow 0} \int e^{i\langle x,\xi\rangle - \frac{\varepsilon}{2}|\xi|^2} \hat{f}(\xi)\, d\xi \\
&= \int_{\mathbb{R}^d}\left(\int_{\mathbb{R}^d} e^{-i\langle x,\xi\rangle - \frac{\varepsilon}{2}|\xi|^2} e^{-i\langle y,\xi\rangle} f(y)\, d^d y\right) d^d\xi
\end{aligned}$$

Making use of the factor of $e^{-\frac{\varepsilon}{2}|\xi|^2}$ allows us to use Fubini's theorem to obtain

$$= \int_{\mathbb{R}^d}\left(\int_{\mathbb{R}^d} e^{i\langle x,\xi\rangle - \frac{\varepsilon}{2}|\xi|^2} e^{-i\langle y,\xi\rangle}\, d^d\xi\right) f(y)\, d^d y.$$

Evaluating the inner integral, we substitute $\eta = \sqrt{\varepsilon}\xi$ to obtain

$$\begin{aligned}
\int_{\mathbb{R}^d} e^{-\langle y-x,\xi\rangle} e^{-\frac{\varepsilon}{2}|\xi|^2}\, d^d\xi &= \varepsilon^{-d/2} \int_{\mathbb{R}^d} e^{-i\left(\left\langle \frac{y-x}{\sqrt{\varepsilon}},\eta\right\rangle\right)} e^{-\frac{|\eta|^2}{2}}\, d^d\eta \\
&= \varepsilon^{-d/2}\mathcal{F}\left[\eta\mapsto e^{-\frac{|\eta|^2}{2}}\right]\left(\frac{y-x}{\sqrt{\varepsilon}}\right) \\
&= (\varepsilon)^{-d/2}(2\pi)^{d/2} e^{-\frac{|y-x|^2}{2\varepsilon}} \\
&= (2\pi)^d\left((2\pi)^{-d/2} e^{-\frac{|y-x|^2}{2\varepsilon}}\right).
\end{aligned}$$

Rewriting as $\varphi_{x,\varepsilon}$, we have

$$\begin{aligned}
\mathcal{G}\circ\mathcal{F}[f](x) &= (2\pi)^d \lim_{\varepsilon\searrow 0} \int_{\mathbb{R}^d} f(y)\varphi_{x,\varepsilon}(y)\, d^d y \\
&= (2\pi)^d \delta_x^d(f) \\
&= (2\pi)^d f(x).
\end{aligned}$$

□

Theorem (Plancherel's Theorem)**:** For any $f, g \in \mathcal{S}\left(\mathbb{R}^d, \mathbb{C}\right)$, one has

$$\int_{\mathbb{R}^d} \overline{\hat{f}(\xi)}\hat{g}(\xi)\, d^d\xi = (2\pi)^d \int \overline{f(x)}g(x)\, d^d x.$$

Proof. The left-hand side can be written as follows

$$\int_{\mathbb{R}^d} \overline{\left(\int_{\mathbb{R}^d} e^{-i\langle x,\xi\rangle} f(x)\, d^d x\right)}\hat{g}(\xi)\, d^d\xi = \int_{\mathbb{R}^d}\int_{\mathbb{R}^d} e^{i\langle x,\xi\rangle}\overline{f(x)}\hat{g}(\xi)\, d^d\xi\, d^d x \tag{$*$}$$

This follows from the estimate with Tonelli's Theorem

$$\int_{\mathbb{R}^d}\int_{\mathbb{R}^d}\left|e^{i\langle x,\xi\rangle}\overline{f(x)}\hat{g}(\xi)\right| d^d x\, d^d\xi = \left(\int_{\mathbb{R}^d}|f(x)|\, d^d x\right)\left(\int_{\mathbb{R}^d}|\hat{g}(\xi)|\, d^d\xi\right)$$

and using the fact that functions on the Schwartz space are integrable. Yet again using Fubini's theorem on $(*)$, we have

$$\begin{aligned}\int_{\mathbb{R}^d}\int_{\mathbb{R}^d} e^{i\langle x,\xi\rangle}\overline{f(x)}\hat{g}(\xi)\, d^d\xi\, d^d x &= \int_{\mathbb{R}^d}\overline{f(x)}\left(\int_{\mathbb{R}^d} e^{i\langle x,\xi\rangle}\hat{g}(\xi)\, d^d\xi\right) d^d x \\ &= \int_{\mathbb{R}^d}\overline{f(x)}\mathcal{G}(\mathcal{F}[g])(x)\, d^d x \\ &= (2\pi)^d\int_{\mathbb{R}^d}\overline{f(x)}g(x)\, d^d x.\end{aligned}$$

□

We note that one can show that $\mathcal{U} = (2\pi)^{-d/2}\mathcal{F}$ is an isometry on the Schwartz space $\mathcal{S}\left(\mathbb{R}^d, \mathbb{C}\right)$. Since $\mathcal{S}\left(\mathbb{R}^d, \mathbb{C}\right)$ is dense in $L^2\left(\mathbb{R}^d\right)$, continuity gives that there is a unique extension $\widetilde{\mathcal{U}}\colon L^2\left(\mathbb{R}^d\right) \to L^2\left(\mathbb{R}^d\right)$. The map $\widetilde{\mathcal{U}}$ is the definition of the Fourier transform on $L^2\left(\mathbb{R}^d\right)$.

Convolution and Properties of the Fourier Transform on $\mathcal{S}\left(\mathbb{R}^d, \mathbb{C}\right)$

Definition: We define the space $\mathcal{O}_M\left(\mathbb{R}^d, \mathbb{K}\right)$ to consist of all functions $f\colon \mathbb{R}^d \to \mathbb{K}$ such that f is C^∞ and for every $\alpha \in \mathbb{N}^d$, there exists $k \in \mathbb{N}$ and a constant $C \geq 0$ such that for all $x \in \mathbb{R}^d$, one has

$$\left|\partial^\alpha f(x)\right| \leq C\langle x\rangle^k.$$

These are known as the smooth functions of tempered growth.

Proposition: Let $g \in \mathcal{O}_M$. Then, for all $f \in \mathcal{S}\left(\mathbb{R}^d\right)$, $fg \in \mathcal{S}\left(\mathbb{R}^d\right)$, and one has that the map $f \mapsto fg$ is continuous.

Proof. We will assume that $|\partial^\alpha g(x)| \leq C\langle x\rangle^{k_\alpha}$.

Now, we have

$$\partial^\alpha(fg) = \sum_{\substack{\beta\in\mathbb{N}^d \\ \beta\leq\alpha}} \frac{\alpha!}{\beta!(\alpha-\beta)!}\partial^{\alpha-\beta}f\,\partial^\beta f.$$

So,

$$\langle x\rangle^k\left|\partial^\alpha(fg)(x)\right| \leq \langle k\rangle \sum_{\substack{\beta\in\mathbb{N}^d \\ \beta\leq\alpha}} \frac{\alpha!}{\beta!(\alpha-\beta!)}\left|\partial^{\alpha-\beta}f(x)\right| C_\beta\langle x\rangle^{k_\beta}$$

$$\leq \sum_{\substack{\beta\in\mathbb{N}^d \\ \beta\leq\alpha}} \frac{\alpha!}{\beta!(\alpha-\beta)!} C_\beta \|f\|_{\alpha-\beta,k+k_\beta}$$
$$< \infty.$$

So, one has that $\|fg\|_{\alpha,k} < \infty$ for all $\alpha \in \mathbb{N}^d$ and all $k \in \mathbb{N}$. The seminorm criterion also gives continuity of the map $f \mapsto fg$. □

Remark: There is a converse to this theorem. Namely, if $g\colon \mathbb{R}^d \to \mathbb{K}$ is a function such that $fg \in \mathcal{S}\left(\mathbb{R}^d\right)$ for all $f \in \mathcal{S}\left(\mathbb{R}^d\right)$.

Bumpfunctionology (for lack of a better term) gives smoothness of such a g, but showing the polynomial bound requires a bit more work.

Example: Monomials $x \mapsto x^\alpha$ satisfy

$$\partial^\beta x^\alpha = \chi_{\{\beta\leq\alpha\}} \frac{\alpha!}{(\alpha-\beta)!} x^{\alpha-\beta},$$

so since $\left|x_j\right| \leq |x| \leq \langle x\rangle$, one has

$$\left|\partial^\beta x^\alpha\right| \leq C_\beta \langle x\rangle^{k_\beta},$$

where $k_\beta = \max\left(0, \left|\alpha-\beta\right|\right)$, and $C_\beta = \chi_{\{\beta\leq\alpha\}} \frac{\alpha!}{(\alpha-\beta)!}$.

We use

$$m_\alpha\colon \mathcal{S}\left(\mathbb{R}^d, \mathbb{K}\right) \to \mathcal{S}\left(\mathbb{R}^d, \mathbb{K}\right)$$
$$f \mapsto x^\alpha f.$$

Proposition: The operator $\partial^\alpha\colon \mathcal{S}\left(\mathbb{R}^d, \mathbb{K}\right) \to \mathcal{S}\left(\mathbb{R}^d, \mathbb{K}\right)$ taking $f \mapsto \partial^\alpha f$ is linear and continuous.

Proof. To be written. □

Theorem (Intertwining Property of Fourier Transform)**:** As continuous linear operators on $\mathcal{S}\left(\mathbb{R}^d, \mathbb{C}\right)$, one has

$$\partial^\alpha \mathcal{F} = (-i)^{|\alpha|} \mathcal{F} \circ m_\alpha$$
$$m_\alpha \circ \mathcal{F} = (-i)^{|\alpha|} \mathcal{F} \circ \partial^\alpha$$

Proof. Using dominated convergence, we may bring the derivative inside the integral, so that

$$\begin{aligned}
\partial_\xi^\alpha \hat{f}(\xi) &= \partial_\xi^\alpha \int_{\mathbb{R}^d} e^{-i\langle x,\xi\rangle} f(x)\, d^d x \\
&= \int_{\mathbb{R}^d} \partial_\xi^\alpha \left(e^{-i\langle x,\xi\rangle}\right) f(x)\, d^d x \\
&= \int_{\mathbb{R}^d} (-ix)^{|\alpha|} e^{-i\langle x,\xi\rangle} f(x)\, d^d x \\
&= (-i)^{|\alpha|} \int_{\mathbb{R}^d} e^{-i\langle x,\xi\rangle} x^\alpha f(x)\, d^d x \\
&= (-i)^{|\alpha|} (\mathcal{F} \circ m_\alpha[f])(x).
\end{aligned}$$

For the other direction, integration by parts yields

$$\xi^\alpha \hat{f}(\xi) = \int_{\mathbb{R}^d} \xi^\alpha e^{-i\langle x,\xi\rangle} f(x)\, d^d x$$

$$
\begin{aligned}
&= \int_{\mathbb{R}^d} i^{|\alpha|} \partial_x^\alpha \left(e^{-i\langle x,\xi\rangle} \right) f(x)\, d^d x \\
&= (-i)^{|\alpha|} \int_{\mathbb{R}^d} e^{-i\langle x,\xi\rangle} \partial_x^\alpha (f(x))\, d^d x \\
&= (-i)^{|\alpha|} (\mathcal{F} \circ \partial^\alpha [f])(x)
\end{aligned}
$$

□

Proposition: Pointwise multiplication $\mathcal{S}\left(\mathbb{R}^d, \mathbb{C}\right) \times \mathcal{S}\left(\mathbb{R}^d, \mathbb{C}\right) \to \mathcal{S}\left(\mathbb{R}^d, \mathbb{C}\right)$ is bilinear and (jointly) continuous.

Proof. We have

$$
\partial^\alpha (fg) = \sum_{\substack{\beta \in \mathbb{N}^d \\ \beta \leq \alpha}} \frac{\alpha!}{\beta!(\alpha-\beta)!} \partial^{\alpha-\beta} f\, \partial^\beta g
$$

so

$$
\begin{aligned}
\langle x \rangle^k \left| \partial^\alpha (fg)(x) \right| &\leq \sum_{\substack{\beta \in \mathbb{N}^d \\ \beta \leq \alpha}} \frac{\alpha!}{\beta!(\alpha-\beta)!} \|f\|_{\alpha-\beta,k} \|g\|_{\beta,0} \\
&\leq \tau_1(f) \tau_2(g)
\end{aligned}
$$

where

$$
\begin{aligned}
\tau_1(f) &= \left(\sum_{\substack{\beta \in \mathbb{N}^d \\ \beta \leq \alpha}} \max_{\beta \leq \alpha} \frac{\alpha!}{\beta!(\alpha-\beta)!} \right) \max_{\beta \leq \alpha} \|f\|_{\beta,k} \\
\tau_2(g) &= \max_{\beta \leq \alpha} \|g\|_{\beta,0}.
\end{aligned}
$$

Therefore, by the seminorm criterion for continuity of multilinear maps, it follows that $(f,g) \mapsto fg$ is continuous. □

Definition: If $f, g \in \mathcal{S}\left(\mathbb{R}^d, \mathbb{K}\right)$, then

$$
(f * g)(x) = \int_{\mathbb{R}^d} f(y) g(x-y)\, d^d y
$$

is called the convolution of two functions.

Note that

$$
\int_{\mathbb{R}^d} \left| f(y) g(x-y) \right| d^d y \leq \int_{\mathbb{R}^d} \langle y \rangle^{-(d+1)} \|f\|_{0,d+1} \|g\|_{0,0}\, d^d y,
$$

so this is well-defined.

Proposition: The convolution $f * g$ is C^∞, and for every α, $\partial^\alpha (f * g) = f * (\partial^\alpha g)$.

Proof. By induction on $|\alpha|$, we start by observing that $\left| \partial_i (f(y) g(x-y)) \right| = \left| f(y) \partial_i (g(x-y)) \right|$, where ∂_i denotes a single partial derivative with respect to the coordinate x_i. Since the derivative of any Schwartz function is Schwartz, and the product of any two Schwartz functions is Schwartz, it follows that the latter function is Schwartz, hence in L^1. Therefore, by Dominated Convergence, it follows that

$$
\partial_i \int f(y) g(x-y)\, d^d y = \int f(y) \partial_i (g(x-y))\, d^d y.
$$

By Clairaut's Theorem, we may write any n-fold partial derivative ∂_x^α as $\partial_i \partial_x^\beta$, where β is a d-dimensional

multi-index satisfying $\left|\beta\right| = |\alpha| - 1$. Since

$$\left|\partial_x^\alpha (f(y)g(x-y))\right| = \left|\partial_i\left(\partial_x^\beta (f(y)g(x-y))\right)\right|,$$

and once again the latter function is a Schwartz function since the derivative of a Schwartz function is Schwartz. Therefore, by Dominated Convergence, it follows that we may exchange derivative and integral, yielding

$$\begin{aligned}
\partial^\alpha (f*g)(x) &= \partial_x^\alpha \int_{\mathbb{R}^d} f(y)g(x-y)\, d^d y \\
&= \partial_i \partial_x^\beta \int_{\mathbb{R}^d} f(y)\partial_x^\beta (g(x-y))\, d^d y \\
&= \partial_i \int_{\mathbb{R}^d} f(y)\left(\partial_x^\beta g\right)(x-y)\, d^d y \\
&= \int f(y)\partial_i \partial_x^\beta (g(x-y))\, d^d y \\
&= \int f(y)\partial^\alpha (g(x-y))\, d^d y \\
&= f*(\partial^\alpha g).
\end{aligned}$$

□

Lemma: For all $u, v \in \mathbb{R}^d$, one has

Proof. We have

$$\begin{aligned}
\langle u+v\rangle^2 &= 1 + |u+v|^2 \\
&\leq 1 + |u|^2 + |v|^2 + 2|u||v| \\
&\leq 1 + 2|u|^2 + 2|v|^2 \\
&\leq 2 + 2|u|^2 + 2|v|^2 \\
&\leq 2\left(1+|u|^2\right)\left(1+|v|^2\right).
\end{aligned}$$

□

Theorem: The convolution $\mathcal{S}\left(\mathbb{R}^d, \mathbb{K}\right) \times \mathcal{S}\left(\mathbb{R}^d, \mathbb{K}\right) \to \mathcal{S}\left(\mathbb{R}^d, \mathbb{K}\right)$ is well-defined, bilinear, and continuous.

Proof. We have that $\partial^\alpha (f*g)$ is C^∞. Taking brackets, we get

$$\begin{aligned}
\langle x\rangle^k \left|\partial^\alpha (f*g)\right| &= \langle x\rangle^k \left|\int_{\mathbb{R}^d} f(y)(\partial_x^\alpha g)(x-y)\, d^d y\right| \\
&\leq \langle x\rangle^k \int \langle y\rangle^\ell \|f\|_{0,\ell} \langle x-y\rangle^m \|g\|_{\alpha,m}\, d^d y,
\end{aligned}$$

where ℓ and m are to be determined. Using the fact that $\langle x\rangle \leq \sqrt{2}\langle y\rangle\langle x-y\rangle$, we get

$$\langle x\rangle^k \int \langle y\rangle^\ell \|f\|_{0,\ell} \langle x-y\rangle^m \|g\|_{\alpha,m}\, d^d y \leq 2^{k/2}\|f\|_{0,\ell}\|g\|_{\alpha,m} \int_{\mathbb{R}^d} \langle y\rangle^{k-\ell} \langle x-y\rangle^{k-m}\, d^d y.$$

Therefore, if we take $m = k$ and $\ell = d + k + 1$, we obtain that the right hand side is a constant, so that the convolution is in Schwartz space. Furthermore, we have shown that $\|f*g\|_{\alpha,k} \leq K\|f\|_{0,\ell}\|g\|_{\alpha,m}$ for some constant $K \geq 0$, so that the map is bilinear and continuous. □

Proposition: Convolution on $\mathcal{S}\left(\mathbb{R}^d, \mathbb{K}\right)$ satisfies

(i) $f*g = g*f$;

(ii) $\partial(f*g) = f*(\partial^\alpha g)$;

(iii) $\partial^\alpha(f*g) = (\partial^\alpha f)*g$.

Proof.

(i) Taking a substitution of $z = x-y$ has Jacobian determinant equal to 1, so that

$$\begin{aligned} f*g(x) &= \int_{\mathbb{R}^d} f(y)g(x-y)\,d^d y \\ &= \int_{\mathbb{R}^d} f(x-z)g(z)\,d^d z \\ &= g*f. \end{aligned}$$

(ii) We showed this earlier.

(iii) By commutativity, we have

$$\begin{aligned} \partial^\alpha(f*g) &= \partial^\alpha(g*f) \\ &= g*(\partial^\alpha f) \\ &= (\partial^\alpha f)*g. \end{aligned}$$

□

Theorem: For all $f, g \in \mathcal{S}\left(\mathbb{R}^d, \mathbb{C}\right)$, one has

$$\mathcal{F}[f*g] = \mathcal{F}[f]\mathcal{F}[g].$$

Proof. Using Fubini's Theorem, we have

$$\mathcal{F}[f*g](\xi) = \int_{\mathbb{R}^d} e^{-i\langle x,\xi\rangle} \int_{\mathbb{R}^d} f(y)g(x-y)\,d^d y\,d^d x.$$

The proper estimate is found by using Tonelli's theorem so that

$$\begin{aligned} \int_{\mathbb{R}^d}\int_{\mathbb{R}^d} \left|f(y)g(x-y)\right| d^d y\,d^d x &= \int_{\mathbb{R}^d} \left|f(y)\right| \int_{\mathbb{R}^d} \left|g(x-y)\right| d^d x\,d^d y \\ &= \left(\int_{\mathbb{R}^d} \left|f(y)\right| d^d y\right)\left(\int_{\mathbb{R}^d} g(z)\,d^d z\right) \\ &< \infty. \end{aligned}$$

Therefore, Fubini's Theorem and the substitution $z = x-y$ gives

$$\begin{aligned} \mathcal{F}[f*g](\xi) &= \int_{\mathbb{R}^d}\int_{\mathbb{R}^d} e^{-i\langle \xi,y\rangle} e^{-i\langle \xi,x-y\rangle} f(y)g(x-y)\,d^d x\,d^d y \\ &= \left(\int_{\mathbb{R}^d} e^{-i\langle \xi,y\rangle} f\,d^d y\right)\left(\int_{\mathbb{R}^d} e^{-i\langle \xi,z\rangle} g(z)\,d^d z\right) \\ &= \mathcal{F}[f]\mathcal{F}[g]. \end{aligned}$$

□

Operations and Fourier Transform for $\mathcal{S}\left(\mathbb{R}^d, \mathbb{K}\right)^*$

First, we define the derivative of a distribution $\varphi \in \mathcal{S}\left(\mathbb{R}^d, \mathbb{K}\right)^*$, we want to define $\partial^\alpha \varphi$.

First, we work like physicists. We will write

$$\begin{aligned}\varphi(f) &= \langle f, \varphi\rangle \\ &= \int_{\mathbb{R}^d} \varphi(x) f(x)\, d^d x,\end{aligned}$$

where we consider $\varphi(x)$ as a generalized function associated to φ. We would then consider the "derivative" of the distribution φ by using integration by parts, so that

$$\begin{aligned}\int_{\mathbb{R}^d} \partial^\alpha \varphi(x) f(x)\, d^d x &= (-1)^{|\alpha|} \int_{\mathbb{R}^d} \varphi(x) \partial^\alpha f(x)\, d^d x \\ &= (-1)^{|\alpha|} \langle \partial^\alpha f, \varphi\rangle \\ &=: \langle f, \partial^\alpha(\varphi)\rangle.\end{aligned}$$

From now on, we will write $\mathcal{S} = \mathcal{S}\left(\mathbb{R}^d, \mathbb{K}\right)$ (with $\mathbb{K}$ most likely equal to $\mathbb{C}$) and $\mathcal{S}^* = \mathcal{S}\left(\mathbb{R}^d\right)^*$.

Definition: Given a multi-index $\alpha \in \mathbb{N}^d$, the map $\partial^\alpha \colon \mathcal{S}^* \to \mathcal{S}^*$ takes

$$\varphi \mapsto \left(f \mapsto (-1)^{|\alpha|} \langle \partial^\alpha f, \varphi\rangle\right).$$

Proposition: The map $\partial^\alpha \colon S^* \to S^*$ is continuous with respect to the strong dual topology.

Proof. We have

$$\partial^\alpha_{\mathcal{S}^*} = (-1)^{|\alpha|} \left(\partial^\alpha_{\mathcal{S}}\right)^*$$

by definition. The transpose/adjoint is strong–strong continuous. □

Now, we note that $\mathcal{S} \subseteq \mathcal{S}^*$ by the standard evaluation. If $\iota \colon \mathcal{S} \to \mathcal{S}^*$ is the corresponding injection, then we need $\partial^\alpha_{\mathcal{S}^*} \circ \iota = \iota \circ \partial^\alpha_{\mathcal{S}}$.

Proof. We recall that for any $f \in \mathcal{S}$, the definition of the duality between $\mathcal{S}$ and $\mathcal{S}^*$ is given by

$$\langle g, \iota(f)\rangle = \int g(x) f(x)\, d^d x.$$

We note that integration by parts gives

$$\begin{aligned}(-1)^{|\alpha|} \left\langle \partial^\alpha_{\mathcal{S}}(g), \iota(f)\right\rangle &= (-1)^{|\alpha|} \int_{\mathbb{R}^d} f(x) \partial^\alpha_{\mathcal{S}}(g(x))\, d^d x \\ &= \int \partial^\alpha_{\mathcal{S}}(f(x)) g(x)\, d^d x \\ &= \left\langle g, \iota\left(\partial^\alpha_{\mathcal{S}}(f)\right)\right\rangle\end{aligned}$$

□

Example: Let $d = 1$, and let

$$H(x) = \begin{cases} 1 & x \geq 0 \\ 0 & x < 0. \end{cases}$$

Define a distribution $\varphi \in \mathcal{S}^*$ by taking

$$\langle f, \varphi\rangle = \int_{\mathbb{R}} H(x) f(x)\, dx.$$

Notice that

$$\left|\varphi(f)\right| \leq \|f\|_{0,2} \int \langle x\rangle^{-2}\, dx,$$

so that φ is indeed continuous. We compute the derivative to be

$$\begin{aligned}\langle f, \partial\varphi\rangle &= -\langle f, \varphi\rangle \\ &= -\int_{-\infty}^{\infty} H(x) f'(x)\,dx \\ &= -\int_{0}^{\infty} f'(x)\,dx \\ &= f(0) - f(\infty) \\ &= f(0) \\ &= \langle f, \delta_0\rangle \\ &=: \underbrace{\int_{-\infty}^{\infty} \delta(x) f(x)\,dx}_{\text{physics notation}}.\end{aligned}$$

That is, we may write $\frac{d}{dx}(H(x)) = \delta(x)$ if we were physicists.

Example: We want now to find the second derivative. That is,

$$\begin{aligned}\langle f, \partial\delta\rangle &= -\langle f', \delta\rangle \\ &= -f'(0) \\ &= \underbrace{\int_{-\infty}^{\infty} \frac{d}{dx}(\delta(x)) f'(x)\,dx}_{\text{physics notation}}\end{aligned}$$

Next, we concern ourselves with multiplication of distributions by elements of $\mathcal{O}_M$.

Definition: If $\varphi \in \mathcal{S}^*$ and $g \in \mathcal{O}_M$, then we define the pointwise product $g\varphi$ by setting it to be the unique distribution such that

$$\langle g\varphi, f\rangle = \langle \varphi, gf\rangle.$$

for all $f \in \mathcal{S}$.

Note that this function is well-defined since $gf \in \mathcal{S}$ whenever $f \in \mathcal{S}$ and $g \in \mathcal{O}_M$. Notice that this is a transpose of the continuous map $f \mapsto gf \mapsto \langle \varphi, gf\rangle$, so that the map taking $\varphi \mapsto g\varphi$ is linear and continuous.

Remark: The set $\iota(\mathcal{S}) \subseteq \mathcal{S}^*$ is sequentially dense, so in particular, the derivative $\partial^\alpha_{\mathcal{S}^*}$ and pointwise product $\varphi \mapsto g\varphi$ are the unique extensions to $\mathcal{S}^*$ from their corresponding operations on $\mathcal{S}$.

Finally, we concern ourselves with Fourier transforms of elements of $\mathcal{S}^*$.

Heuristically,[I] we write

$$\langle f, \varphi\rangle = \int f(x)\varphi(x)\,dx.$$

Imagine we may construct some $\hat{\varphi}(\xi)$ as

$$\hat{\varphi}(\xi) = \int_{\mathbb{R}} e^{-i\langle x,\xi\rangle}\varphi(x)\,d^d x,$$

and "define" the distribution $\hat{\varphi}$ by

$$\langle f, \hat{\varphi}\rangle = \int \hat{\varphi}(\xi) f(\xi)\,d^d\xi.$$

[I] That is, completely and comically unrigorously.

Plugging in directly would then yield

$$\begin{aligned}\int_{\mathbb{R}^d} \hat{\varphi}(\xi) f(\xi)\, d^d\xi &= \int \int e^{-i\langle x,\xi\rangle} \varphi(x) f(\xi)\, d^d x\, d^d\xi \\ &= \int \varphi(x) \int e^{-i\langle x,\xi\rangle} f(\xi)\, d^d\xi\, d^d x \\ &= \left\langle \hat{f}, \varphi \right\rangle \\ &=: \langle f, \mathcal{F}[\varphi]\rangle.\end{aligned}$$

Definition: If $\varphi \in \mathcal{S}^*$, we define the Fourier transform of φ by

$$\langle f, \mathcal{F}[\varphi]\rangle = \langle \mathcal{F}[f], \varphi\rangle.$$

This map extends $\mathcal{F}$ on $\mathcal{S} \subseteq \mathcal{S}^*$. Notice that on $\mathcal{S}^*$, $\mathcal{F}$ is the adjoint of $\mathcal{F}$ on $\mathcal{S}$, so that $\mathcal{F}$ is continuous. Furthermore, exercising the same process on $\mathcal{F}^{-1}$ yields that $\mathcal{F}$ is a strong topological isomorphism of $\mathcal{S}^*$ onto itself.

Remark: The operator $(2\pi)^{-d/2}\mathcal{F}_{\mathcal{S}^*}|_{L^2(\mathbb{R}^d)}$, where we view $L^2\left(\mathbb{R}^d\right) \subseteq \mathcal{S}^*$, is the Fourier transform on L^2, and it is a unitary.

Example: Physicists often write

$$\int_{-\infty}^{\infty} e^{-ix\xi}\, dx = 2\pi\delta(\xi).$$

This makes no sense. We may actually interpret this as $\mathcal{F}[\varphi]$ where φ is the distribution given by the constant function 1. That is,

$$\langle f, \varphi\rangle = \int_{-\infty}^{\infty} f(x)\, dx.$$

If we take the Fourier transform of φ under this notion, we expect it to equal $2\pi\delta(\xi)$. Computing is then given by

$$\begin{aligned}\langle f, \mathcal{F}[\varphi]\rangle &= \langle \varphi, \mathcal{F}[f]\rangle \\ &= \int_{-\infty}^{\infty} \hat{f}(x)\, dx\end{aligned}$$

Multiplying by 1 twice over yields

$$\begin{aligned}&= 2\pi\left(\frac{1}{2\pi}\int_{-\infty}^{\infty} e^{i(0)x}\hat{f}(x)\, dx\right) \\ &= 2\pi f(0) \\ &= \langle 2\pi\delta, f\rangle.\end{aligned}$$

Example: Given some $0 < \alpha < d$, physicists may write

$$\int_{\mathbb{R}^d} e^{-i\langle x,\xi\rangle} \frac{1}{|x|^\alpha}\, d^d x = \frac{\Gamma\left(\frac{d-\alpha}{2}\right)}{\Gamma\left(\frac{\alpha}{2}\right)} 2^{d-\alpha}\pi^{d/2} \frac{1}{|\xi|^{d-\alpha}}.$$

Once again, this doesn't make much sense (i.e., it's not a well-defined Lebesgue integral). Similar to the previous case, we will define an honest distribution $\varphi \in \mathcal{S}^*$ and then establish the Fourier transform of this distribution.

Let

$$\langle f, \varphi\rangle = \int_{\mathbb{R}^d \setminus \{0\}} \frac{1}{|x|^\alpha} f(x)\, dx.$$

We claim that this is a convergent Lebesgue integral. This emerges from the fact that

$$\begin{aligned}\int_{\mathbb{R}^d\setminus\{0\}} \frac{1}{|x|^\alpha}|f(x)|\,d^dx &= \int_{0<|x|\le 1} \frac{1}{|x|^\alpha}|f(x)|\,d^dx + \int_{|x|\ge 1} \frac{1}{|x|^\alpha}|f(x)|\,d^dx \\ &\le \|f\|_{0,0}\frac{2\pi^{d/2}}{\Gamma(d/2)}\int_0^1 r^{d-1-\alpha}\,dr + \|f\|_{0,d+1}\int_{|x|\ge 1}\langle x\rangle^{-(d+1)}\,d^dx \\ &< \infty.\end{aligned}$$

This inequality automatically gives that φ is an honest distribution.

Therefore, our goal is to compute the Fourier transform of φ. We start by observing that, by taking $u = t|\xi|^2$, we have that

$$\begin{aligned}\int_0^\infty e^{-t|\xi|^2}t^{\alpha/2-1}\,dt &= \int_0^\infty e^{-u}\left(\frac{1}{|\xi|^2}\right)^{\alpha/2-1}u^{\alpha/2-1}\,du \\ &= \frac{1}{|\xi|^\alpha}\int_0^\infty e^{-u}u^{\alpha/2-1}\,du \\ &= \frac{1}{|\xi|^\alpha}\Gamma\left(\frac{\alpha}{2}\right).\end{aligned}$$

Furthermore, we compute that

$$\begin{aligned}\mathcal{F}\left[e^{-a|x|^2}\right](\xi) &= \int_{\mathbb{R}^d} e^{-i\langle\xi,x\rangle - a|x|^2}\,d^dx \\ &= (2a)^{-d/2}\int e^{-i\frac{\langle u,\xi\rangle}{\sqrt{2a}}-|u|^2/2}\,d^du \\ &= (2a)^{-d/2}(2\pi)^{d/2}e^{-\frac{1}{2}\left(\frac{|\xi|^2}{2a}\right)} \\ &= \left(\frac{\pi}{a}\right)^{d/2}e^{-\frac{|\xi|^2}{4a}}.\end{aligned}$$

By the definition of $\mathcal{F}[\varphi]$, we have

$$\begin{aligned}\langle f,\mathcal{F}[\varphi]\rangle &= \langle\mathcal{F}[f],\varphi\rangle \\ &= \int_{\mathbb{R}^d\setminus\{0\}}\frac{1}{|\xi|^\alpha}\hat{f}(\xi)\,d^d\xi \\ &= \int_{\mathbb{R}^d\setminus\{0\}}\left(\frac{1}{\Gamma\left(\frac{\alpha}{2}\right)}\int_{\mathbb{R}^d}e^{-t|\xi|^2}t^{\alpha/2-1}\,dt\right)\hat{f}(\xi)\,d^d\xi \\ &= \frac{1}{\Gamma(\alpha/2)}\int_{\mathbb{R}^d\setminus\{0\}}\int_{\mathbb{R}^d}\hat{f}(\xi)e^{-t|\xi|^2}t^{\alpha/2-1}\,dt\,d^d\xi\end{aligned}$$

Observe that

$$\int_{\mathbb{R}^d}\int_0^\infty\left|\frac{1}{\Gamma(\alpha/2)}e^{-t\xi^2}t^{\alpha/2-1}\hat{f}(\xi)\right|dt\,d^d\xi \le \int_{\mathbb{R}^d\setminus\{0\}}\frac{1}{|\xi|^\alpha}\left|\hat{f}(\xi)\right|d^d\xi,$$

so the integral is finite by Tonelli's theorem. By Fubini's theorem, we may switch the order of the integrand, giving

$$= \frac{1}{\Gamma(\alpha/2)}\int_0^\infty\int_{\mathbb{R}^d\setminus\{0\}}\hat{f}(\xi)e^{-t|\xi|^2}t^{\alpha/2-1}\,d^d\xi\,dt$$

Introducing a set of measure zero gives

$$= \int_0^\infty\left(\int_{\mathbb{R}^d}e^{-t|\xi|^2}\hat{f}(\xi)\,d^d\xi\right)\frac{t^{\alpha/2-1}}{\Gamma(\alpha/2)}\,dt$$

$$= \int_0^\infty \left(\int_{\mathbb{R}^d} \overline{e^{-t|\xi|^2}} \hat{f}(\xi)\, d^d\xi \right) \frac{t^{\alpha/2-1}}{\Gamma(\alpha/2)}\, dt$$
$$= \int_0^\infty \left(\int_{\mathbb{R}^d} \overline{\mathcal{F}\left[(4t\pi)^{-d/2} e^{-\frac{|x|^2}{4t}} \right](\xi)} \hat{f}(\xi)\, d^d\xi \right) \frac{t^{\alpha/2-1}}{\Gamma(\alpha/2)}\, dt$$

By Plancherel's Theorem, we get

$$= \int \left((2\pi)^d \int_{\mathbb{R}^d} (4t\pi)^{-d/2} e^{-\frac{|x|^2}{4t}} f(x)\, d^d x \right) \frac{t^{\alpha/2-1}}{\Gamma(\alpha/2)}\, dt$$
$$= \frac{\pi^{d/2}}{\Gamma(\alpha/2)} \int_0^\infty \int_{\mathbb{R}^d} t^{\alpha/2-1-d/2} e^{-\frac{|x|^2}{4t}} f(x)\, d^d x\, dt$$

By Fubini's Theorem and deleting a measure zero set, we get

$$= \frac{\pi^{d/2}}{\Gamma(\alpha/2)} \int_{\mathbb{R}^d \setminus \{0\}} \left(\int_0^\infty e^{-\frac{|x|^2}{4t}} t^{\frac{\alpha}{2}-\frac{d}{2}-1}\, dt \right) f(x)\, d^d x.$$

Using $v = |x|^2/4t$, we get

$$= \frac{\pi^{d/2}}{\Gamma(\alpha/2)} \int_{\mathbb{R}^d \setminus \{0\}} \left(|x|^{\alpha-d} 2^{d-\alpha} \int_0^\infty v^{\frac{d-\alpha}{2}-1} e^{-v}\, dv \right) f(x)\, d^d x$$
$$= \int_{\mathbb{R}^d \setminus \{0\}} \left(\frac{\pi^{d/2} 2^{d-\alpha} \Gamma\left(\frac{d-\alpha}{2}\right)}{\Gamma(\alpha/2)} \frac{1}{|x|^{d-\alpha}} \right) f(x)\, dx$$
$$= \langle f, \mathcal{F}[\varphi] \rangle.$$

The most important example is $d = 3$ and $\alpha = 2$. Then, $\frac{1}{|\xi|^2}$ maps to $\frac{2\pi^{d/2}\Gamma(1/2)}{\Gamma(1)} \frac{1}{|x|} = \frac{2\pi^2}{|x|}$.

In particular, concerning the Poisson equation $-\Delta\varphi = \rho$, one finds that $\varphi = \mathcal{F}^{-1}\left(\frac{1}{|\xi|^2} \right) * \rho$. Therefore, in three dimensions, we have

$$\varphi(y) = \left(\frac{1}{(2\pi)^3} \frac{2\pi^2}{|x|} \right) * \rho$$
$$= \int_{\mathbb{R}^3 \setminus \{x\}} \frac{1}{4\pi |x-y|} \rho(y)\, d^3 y.$$

Normed Vector Spaces

Our main focus is going to be on Banach spaces.

First, we note that if $V_1, \ldots, V_n, W$ are locally convex topological vector spaces, then $\mathcal{L}_n(V_1, \ldots, V_n; W)$ can be equipped with the topology given by

$$\|M\| = \sup\{\tau(M(x_1, \ldots, x_n)) : \tau \text{ a continuous seminorm on } W, (x_1, \ldots, x_n) \in A_1 \times \cdots \times A_n\}.$$

In particular, for $n = 1$ and $W = \mathbb{C}$, this gives the strong topology on the dual V^*. When $V_1, \ldots, V_n, W$ are normed spaces, then the topology on $\mathcal{L}_n(V_1, \ldots, V_n; W)$ is the operator norm topology.

Proposition: If V and W are normed spaces, and $\mathcal{L}(V, W)$ is equipped with the operator norm topology, then if W is complete, so is $\mathcal{L}(V, W)$ is complete.

The Classic Theorems of Normed Spaces

Theorem: If V and W are Banach spaces, and $T\colon V \to W$ is a surjective continuous linear map, then T is an open map.

Corollary: If $T\colon V \to W$ is a continuous linear bijection of Banach spaces, T is an isomorphism of topological vector spaces.

The Riesz Representation Theorem

Our focus will be on the version of the Riesz Representation Theorem for compact metrizable spaces.

Let $(X, \mathcal{T}_X)$ be a topological space. We will write $\mathcal{B}_X$ for the Borel σ-algebra on X — i.e., $\sigma(\mathcal{T}_X)$. We will assume that X is metrizable. Our goal is to study positive Borel measures on X; that is, $\mu\colon \mathcal{B}_X \to [0, \infty]$.

Definition: Let (X, d) be a metric space. Let $A \subseteq X$. Then, we define the diameter

$$\operatorname{diam}(A) \coloneqq \sup_{x,y \in A} d(x, y).$$

In particular, if $A \subseteq B$, then $\operatorname{diam}(A) \subseteq \operatorname{diam}(B)$.

If $x \in X$ and $A \subseteq X$, then we define the distance from x to A by

$$\operatorname{dist}(x, A) = \inf_{y \in A} d(x, y).$$

Notice that if $x, y \in X$ and $z \in A$, we have that

$$\begin{aligned} \operatorname{dist}(x, A) &\leq d(x, z) \\ &\leq d(x, y) + d(y, z), \end{aligned}$$

so we obtain that for all $x, y \in A$, $\left|\operatorname{dist}(x, A) - \operatorname{dist}(y, A)\right| \leq d(x, y)$, meaning that the distance function is continuous.

Proposition: If A is closed, then $d(x, A) = 0$ if and only if $x \in A$.

Proposition (Urysohn's Lemma for Metric Spaces)**:** Let A and B be nonempty, closed, and disjoint. Then, there exists $f\colon X \to \mathbb{R}$ continuous such that $f(x) \in [0, 1]$, $f|_A = 1$, and $f|_B = 0$.

Proof. Take

$$f = \frac{\operatorname{dist}(x, B)}{\operatorname{dist}(x, A) + \operatorname{dist}(x, B)}.$$

□

Remark: In fact, $f(x) = 1$ if and only if $x \in A$, and similarly, $f(x) = 0$ if and only if $x \in B$.

The notion of bumpfunctionology is a smooth version of the Urysohn lemma.

From now on, assume that X is a nonempty compact metrizable space with compatible metric d. Furthermore, we will assume that $\mathbb{K} = \mathbb{R}$, and so $C(X)$ is the space of real-valued continuous functions on X. Let $\mu\colon \mathcal{B}_X \to [0, \mu(X)]$ be a finite Borel measure on X with $\mu(X) < \infty$.

Define $\mathcal{R}[\mu]\colon C(X) \to \mathbb{R}$ given by

$$\mathcal{R}[\mu](f) = \int_X f \, d\mu.$$

This map is well-defined, $\mathbb{R}$-linear, and continuous. The continuity emerges from the fact that for any $f \in C(X)$, we have

$$\left| \int f \, d\mu \right| \leq \int |f| \, d\mu$$

$$\leq \|f\|\mu(X),$$

so that $\mathcal{R}[\mu] \in C(X)^*$ is continuous of operator norm at most $\mu(X)$. Furthermore, we have

$$\mathcal{R}[\mu](1) = \mu(X),$$

so $\left\|\mathcal{R}[\mu]\right\|_{\text{op}} = \mu(X)$.

Definition: Let $L\colon C(X) \to \mathbb{R}$ be an $\mathbb{R}$-linear form. We say L is positive if for all $f \geq 0$, one has $L(f) \geq 0$.

Proposition: Let $L\colon C(X) \to \mathbb{R}$ be a positive. Then, L is positive if and only if L is continuous and $\|L\|_{\text{op}} = L(1)$.

Proof. Suppose L is positive. Then, $\|f\|1 \leq f \leq \|f\|1$, so that $\|f\|L(1) \leq L(f) \leq \|f\|L(1)$. Therefore, $|L(f)| \leq \|f\|L(1)$. Thus, L is continuous with norm at most $L(1)$; taking $f = 1$ yields the desired outcome.

Let $f \geq 0$. By the hypothesis, for all $g \in C(X)$ we have $|L(g)| \leq \|g\|L(1)$. Note that $0 \leq f \leq \|f\|1$. In particular, this means that $0 \leq \|f\|1 - f \leq \|f\|1$. Take $g = \|f\|1 - f$. Then, we have that $\|g\| \leq \|f\|$. Therefore, we have that $L(g) \leq \|g\|L(1) \leq \|f\|L(1)$. Yet, substituting the definition of f, we have

$$\begin{aligned} L(\|f\|1 - f) &= \|f\|L(1) - L(f) \\ &\leq \|f\|L(1), \end{aligned}$$

so that $L(f) \geq 0$. □

We have a map

$$\mathcal{R}\colon \{\text{finite regular positive Borel measures on } X\} \to \{\text{positive } \mathbb{R}\text{-linear forms on } C(X)\}$$

taking $\mu \mapsto \mathcal{R}[\mu]$.

Theorem (Riesz Representation Theorem)**:** The map $\mathcal{R}$ is a bijection.

We will first show that $\mathcal{R}$ is an injection. We will let (X, d) be a compact metric space. Let μ be a finite Borel measure on X. For any $A \subseteq X$ (not necessarily Borel), define

$$\mu^*(A) = \inf\{\mu(U) : U \text{ open}, A \subseteq U\}.$$

Note that $0 \leq \mu^*(A) \leq \mu(X)$.

Proposition: The map $\mu^*\colon \mathcal{P}(X) \to \mathbb{R}$ is an outer measure.

Proof. Observe that $\emptyset$ is open, so $\mu^*(\emptyset) = 0$.

If $A \subseteq B$, then for any $U \supseteq B$, we have $U \subseteq A$, so that $\mu^*(A) \subseteq \mu^*(B)$.

Finally, suppose $\{A_n : n \in \mathbb{N}\}$ is a sequence of subsets. Let $\varepsilon > 0$. For any $n \in \mathbb{N}$, there is $U_n \supseteq A_n$ such that $\mu(U_n) \leq \mu^*(A_n) + \varepsilon 2^{-(n+1)}$. Let $W = \bigcup_n U_n$. Then,

$$\begin{aligned} \mu^*\left(\bigcup_{n=0}^{\infty} A_n\right) &\leq \mu(W) \\ &\leq \sum_{n=0}^{\infty} \mu(U_n) \\ &\leq \sum_{n=0}^{\infty} \left(\mu^*(A_n) + \varepsilon 2^{-(n+1)}\right) \\ &\leq \sum_{n=0}^{\infty} \mu^*(A_n) + \varepsilon. \end{aligned}$$

Since $\varepsilon > 0$, we obtain our desired result. □

Recall that if $A \subseteq X$ is any subset, then A is μ^*-measurable if and only if for any $E \subseteq P(X)$, we have $\mu^*(E) = \mu^*(E \cap A) + \mu^*(E \cap A^c)$. That is, A is a good "cookie cutter" for μ^*.

Proposition: We have A is μ^*-measurable if and only if $\mu^*(X) = \mu^*(A) + \mu^*(A^c)$.

Proof. We only need to show that $\mu^*(E) \geq \mu^*(E \cap A) + \mu^*(E \cap A^c)$ assuming this property.

Let $U, V, W \subseteq X$ be open such that $A \subseteq U$, $A^c \subseteq V$, and $E \subseteq W$, and let $\varepsilon > 0$. Our goal is to show that given such a W, one can pick U and V such that $\mu(W) \geq \mu(W \cap U) + \mu(W \cap V) - \varepsilon$.

Notice that if C and D are Borel, then $\mu(C)+\mu(D) = \mu(C \cup D)+\mu(C \cap D)$. Taking $C = W \cap U$ and $D = W \cap V$, then

$$\begin{aligned}\mu(W \cap U) + \mu(W \cap V) &= \mu(W \cap (U \cup V)) + \mu(W \cap U \cap V) \\ &= \mu(W) + \mu(W \cap U \cap V),\end{aligned}$$

where we use the fact that $A \subseteq U$ and $A^c \subseteq V$ so that $U \cup V = X$. Our amended goal is to find W such that $\mu(W \cap U \cap V) \leq \varepsilon$. In fact, it is enough to pick U and V such that $\mu(U \cap V) \leq \varepsilon$.

To find this, we pick U such that $\mu(U) \leq \mu^*(A) + \varepsilon/2$ and V such that $\mu(V) \leq \mu^*(A^c) + \varepsilon/2$. Then,

$$\begin{aligned}\mu(U \cap V) &= \mu(U) + \mu(V) - \mu(U \cup V) \\ &= \mu(U) + \mu(V) - \mu(X).\end{aligned}$$

Therefore, we have

$$\begin{aligned}\mu(U \cap V) &\leq \mu^*(A) + \varepsilon/2 + \mu^*(A^c) + \varepsilon/2 - \mu(X) \\ &= (\mu^*(A) + \mu^*(A^c) - \mu(X)) + \varepsilon \\ &= \varepsilon.\end{aligned}$$

This yields our desired result. □

Definition: If $A \subseteq X$ is an arbitrary subset, then we define

$$\mu_*(A) = \sup\{\mu(K) : K \text{ compact}, K \subseteq A\}.$$

Proposition: For every $A \subseteq X$, $\mu_*(A) = \mu(X) - \mu^*(A^c)$.

So, A is μ^*-measurable if and only if $\mu^*(A) = \mu_*(A)$.